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REVIEW ARTICLE

The state of the Yamuna River: a detailed review of water quality assessment across the entire course in India

Madhuben Sharma[1] · Sameeksha Rawat[1] · Dheeraj Kumar[1] · Amit Awasthi[2] · Abhijit Sarkar[3] · Atul Sidola[4] · Tanupriya Choudhury[5,6] · Ketan Kotecha[6,7]

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Abstract
The Yamuna River, a vital water source in India, poses a profound challenge concerning water purity across its entire stretch. The comprehensive review aims to thoroughly examine the river's water quality, shedding light on the sources of pollution and their consequences for both ecological systems and public health. The primary objective of this review is to examine the published research papers concerning the Yamuna River water quality stretching from Yamunotri to Prayagraj and its resulting impact on human health. This paper also comprises a wide range of pollutants mainly caused by human activity; during the strange period of COVID-19 lockdown, when all industries were closed, resulting in changes in water quality, signifies the destructive effects of human activity on the river. Studies uncover that the most contaminated areas are Nizamuddin of Delhi region and D/S of Agra in Uttar Pradesh, which includes the foremost level of faecal coliforms to be around 210000–11000000 and 450–6100000, respectively. The total coliforms were found to be between 700000–28000000 and 2200–32000000, respectively. Biochemical oxygen demands, industrial discharge, urban waste and agriculture are identified as the most responsible factors for this contamination. After the COVID-19 lockdown, all industries were open, and now, the conditions are the same as before COVID-19. The primary insight to be assembled is that the ecological balance of the Yamuna River and public health depend on the immediate requirement for effective wastewater treatment solutions. Besides offering valuable data by compiling findings from multiple studies, this review underscores the importance of implementing stringent regulations on industrial emissions, upgrading sewage treatment plants, and promoting eco-friendly farming methods to tackle pollution in the Yamuna River and also manage the rural and urban areas of the sewage pipeline plan. It stresses the importance of safeguarding the Yamuna River ecosystem's inherent socioeconomic benefits while alleviating the environmental harm caused by pervasive pollution. Essentially, the study calls for prompt and comprehensive measures to ensure the sustainable health of this crucial water resource in India.

Keywords Assessment · BOD · COD · Sources · Yamuna River · Water quality

Introduction

The essentiality of freshwater availability cannot be denied, as it aids in the development of society, the economy, politics, and rivers, which are vital to the socioeconomic progress of every nation. However, river systems are extensively contaminated nowadays because of the growing danger to water quality—which is lowered by factors including urbanization, industry, development projects, and human density (Mansor et al. 2024; Akhtar et al. 2021). Pollution adversely affects freshwater quality, primarily due to industrial and agricultural activities (Hothi et al. 2022; Koop and Leeuwen 2017). Figure 1 illustrates several causes of water pollution, such as construction, runoff from farmland, and urban and industrial effluents. This shows how several factors that affect water quality are intricately connected. River water's qualitative and quantitative attributes are significantly impacted by variations in surface water runoff, precipitation, groundwater flow, and human water usage throughout the year (Pattnayak et al. 2023; Kluska and Jabłońska 2024).

Since the Himalayan rivers are essential to life and greatly influence traditions like mass bathing on festivals, they are essential to Indian culture (Haberman 2023). Three principal rivers emerge from the Himalayan highlands: Yamuna, Ganga, and Brahmaputra. These rivers are utilized for transportation, irrigation, fishing, hydroelectric

Extended author information available on the last page of the article

Fig. 1 Sources of water pollution (Anderson et al. 2019)

power production, and cultural purposes. This comprehensive study concentrates on the Yamuna River, which is located at Yamunotri and runs through the states of Madhya Pradesh (MP), Himachal Pradesh (HP), Haryana, Delhi, and Rajasthan.

The need for clean drinking water is growing daily and is a global issue. In the meantime, the two main sources of the problem are the declining state of our environment and the growing concern over waterborne illness (Jaspal and Malviya 2020; Sharma et al. 2020a, b). Industrial areas are the prime area to be concerned about. Numerous contaminants have significantly altered the water quality in these areas, endangering the health of both humans and animals (Koop and Leeuwen 2017). The water quality of the Yamuna River continues to decline downstream into Haryana and Uttar Pradesh as a result of sewage and wastewater discharge from large urban areas like Panipat (drain no. 2), Sonipat (drain no. 6), and Yamuna Nagar (ditch drain) (Rajan and Nandimandalam 2024).

The state of pollution in several river segments caused the decline in the river's water quality because fresh water for drinking and irrigation has been eliminated, and the cumulative release of industrial, residential, and agricultural wastewater into the river has turned the stretch of the river into an open sewer (Kayastha 2024). The health of rivers is at even greater risk due to extreme climate-related events such as record-breaking heat waves and altered precipitation patterns (Pattnayak et al. 2023; Awasthi et al. 2022a, b; Yadav et al.2023). Addressing this requires a multifaceted approach to safeguard the water quality of the Yamuna and potentially prevent the spread of disease, which is crucial to assess the water's quality and locate the sources of pollution, such as agricultural runoff, industrial effluents, untreated sewage discharge, commercial and household wastewater, to halt the spread of illness (Muthaiyah 2020). Determining these sources and assessing the water quality is essential. This study focuses on the Yamuna River since it is one of India's most polluted rivers, which is unsafe to drink or take a bath. Even the contaminated fruits, vegetables, and other foods cultivated in its basin pose a health risk because of the contaminated water. But it seems logical to assume that pollution has become a concern. The problem, however remains challenging even after massive amounts of money have been spent on its solution.

This study is based on a critical review of the literature and primary data collected through different search engines such as Google Scholar and ScienceDirect from 2019 to 2023. The secondary data sources include the water quality data of River Yamuna in the year 2020 from the Central Pollution Control Board (CPCB). These sources were examined to identify the underlying water values driving the water management policies and plans in different stretches of the Yamuna River. This paper is divided into nine sections. Section "River Yamuna" comprehensively explores the different segments of the Yamuna River in India. Section "Sources of contamination in Yamuna River" discusses the sources of contamination in the water body such as industrial effluents, domestic wastewater, pollution from agriculture, biological contamination, and heavy metal pollution sources, and their impact on human health. Section "Water quality assessment" examines the state-wise studies of the River Yamuna, including Uttarakhand, Himachal Pradesh, Uttar Pradesh, Rajasthan, Haryana, Delhi, and Madhya Pradesh. Section "State-wise studies of River Yamuna" contains a discussion on the water quality concerning COVID-19. Section "Observations of water quality with respect to period of COVID-19" discusses the management and treatment of wastewater. The paper ends with concluding remarks and future perspectives. This

review aims to gather information and studies on several contaminants that impact the quality of Yamuna water. Decision-makers will find the information useful in developing strategies to manage water pollution in the Yamuna.

River Yamuna

The Yamuna River, beginning at the Yamunotri glacier in Uttarakhand, serves as the lifeblood of the Himalayan region. Steeped in Hindu mythology and revered for its cultural significance, the Yamuna faces a growing threat like pollution, climate change, and floods, endangering its ecosystem and the well-being of those who rely on it. This critical waterway serves multiple purposes. Homes, industries, and farms all depend on its waters. Power plants, textile mills, paper mills, and chemical factories utilize it for production and cooling. Agriculture, the primary consumer of the Yamuna's water, irrigates vast tracts of land along the river's course. The vast majority of people living in the Himalayas rely exclusively on the river for water. Furthermore, millions in cities like Delhi, Mathura, Agra, and Allahabad quench their thirst and find sustenance thanks to this precious resource. The Yamuna's future health is critical for both the environment and the communities it sustains.

The Yamuna River is the principal tributary of the Ganga in the northern regions of India. It is sometimes referred to as the Jamuna or the Jumna. The percentage distribution of the major tributaries is shown in Fig. 2, with Hindon having the lowest proportion (2%) and Chambal having the largest percentage (40%) of all of them (Mishra et al. 2016). The catchment area of these tributaries is discussed in Table 1. It is a perennial river that receives its water from groundwater, precipitation, and snowmelt runoff (Joshi et al. 2023). One of the major natural sources of rivers is glaciers (Awasthi et al. 2022a, b). The Yamuna River originated from a glacier known as Yamunotri, which is located in the state of UK. At a height of 6387 m, this glacier is located on the western flanks of the Himalayan Banderpoonch peaks. Culturally and spiritually significant, the Yamuna River is home to many pilgrimage sites such as Yamunotri, Mathura, Paonta Sahib, Bateshwar, Vrindavan, and Allahabad. Numerous urban areas along its banks, including Yamunanagar, Sonipat, Gautam Budh Nagar, Delhi, Faridabad, Agra, and Mathura, further highlight its

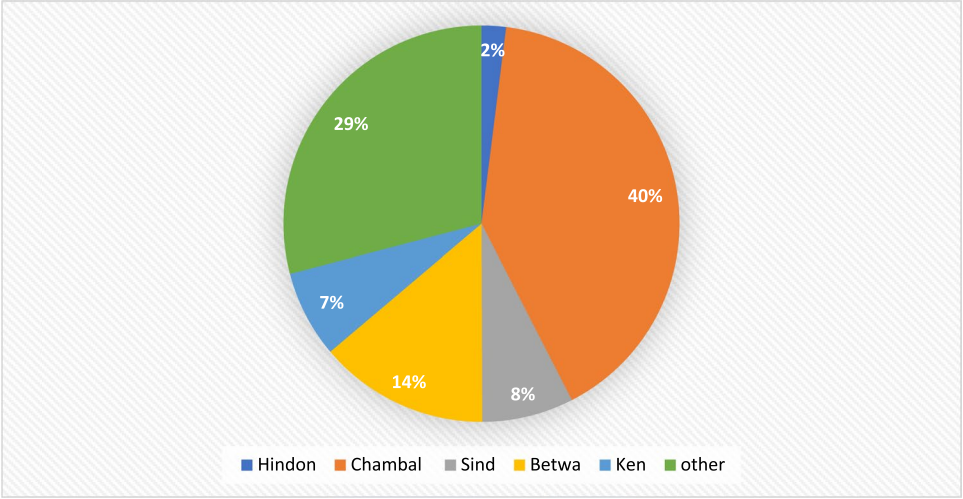

Fig. 2 Major tributaries of River Yamuna

Table 1 State-wise total catchment area, percentage of contribution, drainage basin, percentage of drainage basin (Naithani and Pandel 2015; Upadhyay et al. 2013)

S. no	Location	Total catchment area (km^2)	Drainage basin (km^2)	Percentage drainage basin (%)	Areas in major tributaries (km^2)					
					Chambal	Betwa	Hindon	Ken	Sind	Others
1	UK	3771	21,769	33	–	–	–	–	–	3771
2	UP	70,437	166,231	1	452	14,438	748	3,336	748	44,38
3	HP	5799	2,320	1	–	–	–	–	–	5799
4	Rajasthan	102,883	7422	19	79,495	–	–	–	–	23,388
5	Haryana	21,2883	34,341	11	–	–	–	–	–	21,265
6	Delhi	1485	1485	11	–	–	–	–	–	–
7	MP	14,028	72,700	24	59,838	33,502	–	21,090	25,131	647

importance in daily life. Despite its cultural significance, the river faces challenges from human activities, impacting its fragile ecosystem (Arun et al. 2015).

Table 4 presents variations in the catchment regions of different Indian states, emphasizing the diverse contributions to the drainage basin (Naithani and Pandel 2015). UK and MP play crucial roles with larger catchment areas, while UP, though having a bigger catchment area, contributes relatively less. Smaller catchment regions of HP, Rajasthan, Haryana, and Delhi also make significant contributions to the drainage basin. This diversity underscores the complex dynamics of the Yamuna River and the need for comprehensive management strategies to address its challenges.

Segments of Yamuna River in India

The Yamuna River has been categorized into five distinct segments, namely the Upper, the Himalayan, Delhi, Diluted, and the Eutrophicated Segments which are based on its biological and hydrological characteristics (Sinha 2023). As the river flows through Haryana, specifically reaching Hathnikund/Tajewala, it passes through Ponta Sahib and is directed for irrigation into both the Western Yamuna Canal (WYC) and the Eastern Yamuna Canal (EYC) (Zehra et al. 2023). However, some sections between Tajewala and Delhi experience drying up during the dry season, primarily because there is no downstream water flow to the Tajewala barrage (Sharma and Gupta 2022). This situation changes as the river regains water through groundwater recharge and benefaction from a feeding channel via Som Nadi upstream of Kalanaur. After covering approximately 224 km, the river reaches Delhi near Palla village, where it is tapped once again at Wazirabad via barrage to provide drinking water to Delhi.

Proceeding downstream, there is a second barrage known as Okhla Barrage, located 22 kms after the Wazirabad barrage. The Okhla barrage directs river water towards the Agra Canal for irrigation (DAR 2023). Additionally, water from the Shahdara drain, originating from Noida, East Delhi, and Sahibabad, contributes to the river's flow past the Okhla Barrage (DAR 2023). The journey of the Yamuna concludes at Prayag (Allahabad), where it merges with the Ganga and an underground River Saraswati, receiving water from several significant tributaries (Sharma and Kansal 2011). Table 2 (Sharma and Gupta 2022; Mishra 2010; Sharma 2015) offers a comprehensive overview of the conditions and parameters influencing the quality of water in each section of the Yamuna River.

Sources of contamination in Yamuna River

The 2% of the entire catchment of river Yamuna flows via the National Capital Territory (NCT), Delhi and collects 79% of the river's total pollution in a 48-km stretch resulting in the most polluted section of Yamuna River (Arora et al. 2023). The primary cause of pollution in the river, accounting for around 85%, is domestic sources. These include industrial effluents, raw manure, waste and dead body disposal, idol worship, and contaminants from water used in streams (Parihar et al. 2019).

According to available sources, surface water can be classified in 12 different ways, as illustrated in Fig. 3. The five major locations receiving water from these sources are listed in the centre column of Fig. 3. It is needed to understand the pollution sources to develop effective mitigation techniques. This discussion explores the various factors causing the deterioration of Yamuna's water quality. Each source contributes substantially to the deterioration of the river's health, including biological pollutants, heavy metals (HM), pollution from agriculture, untreated domestic wastewater, and industrial effluents released by industries. In the subsequent sections, each pollution source is discussed in detail, shedding light on the specific pollutants and their impact on the Yamuna's delicate ecosystem.

Industrial effluents

The River Yamuna is also described as the "River of grief " and "dirty river" in Delhi, Agra, and Mathura. This is mainly because of the seepage of untreated wastewater from the majority of Delhi's factories into the river, resulting in water that resembles an industrial sewer (Sarkar et al. 2017; Sarker et al. 2021). Most industrial complexes near riverbanks either do not treat or only partially treat effluents before discharging them into waterways (Tripathi et al. 2008; Syeed et al. 2023). River water is extensively utilized for industrial production, thermal power, and hydroelectricity generation (Syeed et al. 2023; Sarkar et al. 2019). The discharge of industrial wastes leads to elevated concentrations of HM contamination in the area (Yadav and Khandegar 2019).

In cities such as Faridabad, Mathura, Delhi, and Agra, numerous industrial facilities emit substantial quantities of untreated water into the Yamuna (Yadav and Yadav 2024). According to a report 359 industrial facilities dispose of their effluents indirectly or straight into the river. It has been reported that 42 industrial units in Delhi directly contribute to Yamuna contamination (Sharma 2021).

Domestic wastewater

Ninety per cent of the city's residential wastewater enters the Yamuna. It contains chemicals, phosphate compounds,

Table 2 Streams of the River Yamuna (Sharma and Gupta 2022; Mishra 2010; Sharma 2015)

River segments	Segment area	Approx. segment length (km)	BOD (mg/L)	COD (mg/L)	TDS (mg/L)	DO (mg/L)	Description
Himalayan Segment	From the point of its origin until the Tajewala Barrage	172	0–3	6–10	6–10	6–10	Turbulent stream with nearly perfect water quality up to the Tajewala Barrage. On both banks, canals extract water for diverse purposes
Upper Segment	(Upstream of Delhi) Tajewala Barrage to Wazirabad Barrage	224	1–3	10–7	–	10–7	Wastewater and fresh water from rural and urban areas in the basin make up the dry weather flow. Water is kept in storage in Wazirabad to suit Delhi's needs
Delhi Segment (highly polluted)	Wazirabad to Okhla Barrage (Delhi)	22	3–25	7–1	–	7–1	The Wazirabad Barrage does not discharge any water. Delhi-originating wastewater makes up the dry weather flow. Water is sidetracked to the states of Haryana and U.P. for irrigation via the Agra canal thanks to the barrage at Okhla
Eutrophicated (highly polluted)	Okhla Barrage to confluence with Chambal River	490	18–6	–	–	1–12	The dry water flow, which is made up of sewage from urban and rural areas, does not discharge any water from the Okhla barrage
Diluted segment	Chambal confluence to confluence with Ganga River at Allahabad	468	13–1	–	–	11–7	The Chambal water dilutes the river, giving it new vitality. This reach also includes tributaries from M.P. and Rajasthan

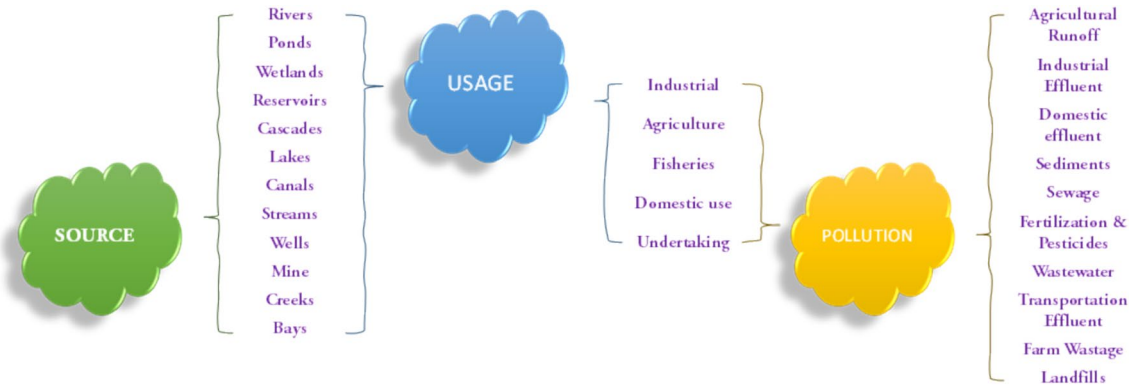

Fig. 3 Categorization of water: sources, use, and possibility of contamination (Syeed et al. 2023)

laundry waste water, detergent, and other materials that cause harmful foam in the river (Vrat 2024). Usha and Singh (2024) reported a phosphate content of 0.51 mg/L which was higher than the standard range of 0.005–0.05 mg/L. The rivers were covered with toxic froth due to the excess phosphate concentration (Usha and Singh 2024).

Pollution from agriculture

Agricultural activities along the river's banks in Delhi contribute to pollution. Crops like radish, cabbage, cauliflower, tomatoes, and spinach are planted along the riverbanks. Runoff from agricultural farms during monsoon and non-monsoon rains, along with the discharge of irrigated water containing residues of insecticides, artificial fertilizer, pesticides, herbicides, and farmyard waste, directly or indirectly damages the river (Sharma et al. 2015). Agriculture is prevalent in the catchment areas and along the entire Yamuna Riverbank, leading to the drying up of river streams during the non-monsoon season and contributing to pesticide residue in the river (Mishra 2010).

Biological contamination

The Yamuna is significantly contaminated throughout its path through the NCT (Delhi), experiencing a conspicuous increase in Biochemical Oxygen Demand (BOD) and organic pollutants. BOD levels, crucial for measuring oxygen needed for microorganisms to break down organic substances, far exceed acceptable limits (Koop and Leeuwen 2017). The Yamuna's total BOD level is approximately 93 mg/L, as compared to the CPCB requirement of 3 mg/L (Patel et al. 2020). In comparison to the World Health Organization (WHO) criterion of 5.0 mg/L, where they represent a hazard to aquatic life due to insufficient oxygen delivery, the obtained values were relatively low.

The Chemical Oxygen Demand (COD) measures both biodegradable and non-biodegradable compounds. Koop and Leeuwen (2017) found that the mean annual COD concentrations in the Yamuna River fluctuated from 202.66 mg/L in the summer to 64 mg/L during the monsoon, 89 mg/L in the winter, and 71.11 mg/L in the spring. These results are relatively low, which aligns with the WHO guidelines limit of 250 mg/L. Lower BOD and COD levels indicate the existence of residential and commercial seepages as well as lower oxygen requirements, all of which point to a greater level of toxicity in the river water. The waterbody also consists of detergents which come under domestic waste (Sharma et al. 2014).

HM pollution sources

Naturally occurring elements are heavy metals. Due to the intricate interactions between human activities and natural processes, it is becoming more prevalent in the natural environment (Rajan and Janardhana Raju 2023). HMs released as a by-product of industrial activities pose a significant threat to water bodies, especially in heavy industrial zones (Syeed et al. 2023). The Yamuna experiences HM contamination, with metals like Cr, Cd, Zn, Cu, Ni, and Pb (Kr et al. 2018). It has been the main concern for science, health, and the environment over the effects of some of the enormously poisonous, detrimental to the environment, and human carcinogenic metals rising availability and/or emission (Okpara et al. 2022). Therefore, water's high content of HM has an impact on the water's quality (Vasistha and Ganguly 2020).

Metal ions are hazardous, risky, and destructive among other organic and inorganic water contaminants because of the way they naturally degrade tissue (Malik 2014). Some harmful HM, such as Chromium (Cr), Mercury (Hg), Cadmium (Cd), Silver (Ag), Lead (Pb), Barium (Ba), Arsenic (As) Thallium (Tl), and Selenium (Se), are included in the metallic compounds (Syeed et al. 2023). HM contamination is categorical by proving its existence in the water environment. Table 3 suggested that HMs may contaminate aquatic ecosystems, emphasizing the need for monitoring and

controlling their presence to safeguard both human health and water quality. River water can include iron from geological sources, such as weathering and the breakdown of iron-rich minerals like magnetite, hematite, and illite, or household and industrial effluents coming from drains (Sarkar and Shekhar 2018). Also, due to mining, leaching from soils or riverbanks, geothermal springs, and anthropogenic sources, the concentration of As is higher in the surface waters globally (Rajan and Nandimandalam 2024). The breakdown of Mn-rich rocks, such as limestone, shale, and mafic rock, as well as human-generated contaminants such as sewage and untreated industrial effluent, are frequently linked to the occurrence of Mn in surface water (Hou et al. 2020).

Significant levels of different heavy metals have been found in the Yamuna River's water, depending on the monitoring of the river's quality. The most prevalent element in the Yamuna water is Fe, even beyond the permissible limit. Various health problems, including inhibited growth and development, organ damage, cancer, impairment of the nervous system, etc., can result from such high concentrations of heavy metals in the water (Jaiswal et al. 2022).

According to (Sehgal et al. 2012), two areas between the Okhla and Wazirabad barrages that appeared to be hotspots for river water degradation were found to have greater concentrations of several HM. Yadav and Khandegar 2019 observed that the broad flow of industrial wastes into the Yamuna River was the cause of the fluctuating concentration of HM. The contamination of HM in the aquatic environment has grown to be a universal concern in recent years due to their invulnerability and the poisonous effects that the majority of them have on species. Ni, Cd, and Pb, three HM, are hardly ever found in considerable concentrations (Nielsen et al. 1999; Malik 2014).

Even at low concentrations, HMs are extremely poisonous and can have negative long-term consequences on the environment as well as human well-being. It can lead to various health issues, including cancer, organ damage, neurological diseases, and developmental anomalies. Large-scale industrial accidents or mining disasters, which involve the release of HM, can have devastating immediate effects that generate media attention and public protest. Therefore, the next section focuses on HM pollution, but it is essential to comprehend that other pollution sources, like industrial effluents, agricultural runoff, etc., also have a substantial effect on human health and water quality.

Impact of HM on human health

HMs are identified to have detrimental consequences on health and have been demonstrated to enter through water and food in the human figure (Malik et al. 2014). Certain HM species, including Ni, Fe, Zn, Mn, and Cu, are required by the human body and other organisms at minimal levels. However, when their maximum concentration limit (MCL) is exceeded, these metals pose a major risk to both the ecosystem and human well-being (Sankhla et al. 2016; Srivastava and Majumder 2008). Table 4 highlights the primary sources of these HMs, the MCL requirements for those metals in water, and the illnesses and symptoms that can result from exposure to those metals (Okpara et al. 2022; Fiyadh et al. 2023). It emphasizes the need to observe and control the quantity of these HMs in water to safeguard human health and address potential health concerns.

Higher levels of Pb are associated with adult peripheral neuropathy and cognitive impairment in children (Kaur and Mehra 2012). Chronic lead exposure may have negative

Table 3 Presence of HM in diverse water conditions (Bhateria and Jain 2016; Kr et al. 2018)

HM	Contaminated marine water	Domestic wastewater	Contaminated Rivers & Lakes	Water affected by industries	Water affected by mining
Cr	✓	✓	✓		✓
Fe		✓			
Ni		✓	✓	✓	
Cu	✓	✓	✓	✓	
Al					✓
Mn		✓			
Co		✓			
Sr				✓	
Zn	✓	✓	✓		
Cd	✓	✓	✓	✓	✓
Pb	✓		✓	✓	✓
U			✓	✓	
Hg			✓	✓	✓

"✓" indicates the presence of HM in water ecosystem. Ni-nickel, Zn-zinc, Cu-copper, Hg-gold

Table 4 HM and their effect on human health (Fiyadh et al. 2023; Okpara et al. 2022)

S. no	HM	Major sources	MCL standard (mg/L)	Symptoms and diseases	References
1	Pb	Pb rich batteries, sanitation PVC pipes, lunch boxes, fuels, electronics parts manufacturing, PVC Pb paints recycling, pigments of painting, processing of metals, mining activities, electroplating, agriculture, leather tanning, Pb batteries, and jewellery	0.06	Damage of CNS, damage in the reproductive systems, liver and kidney diseases, lowering of IQ, attitudinal dysfunctions, reduces bone's growth, stupor, ataxia, and hyperirritability	Liu et al. (2017); Okpara et al. (2022)
2	Cr	Data storage, coasting operations, chemicals, pigments, textiles, metallurgy, wood treatment, leather tanning, and electroplating industries	0.05	Bleedings and irritations in the nose, ulcers and skin rash, mutation of genetic material, debilitated immune system, teratogenicity, kidney & liver impairment, embryotoxicity, dermatitis, toxicity in reproduction, adverse reactions, septum perforation, and causes mutation & cancer,	Dinari and Haghighi (2018); Cao et al. (2018)
3	Cu	Photovoltaic cells, tanning, and fertilizers	0.25	Anaemia, kidney dysfunctions, arthritis, allergies, haemorrhaging, diabetes, adreno-cortical hyperactivity, alopecia, cystic fibrosis, and autism	Selvaraj et al. (2011)
4	Cd	Batteries made up of Cd & Ni, electroplating industries, plastic stabilizers, combustion of coals, synthetic rubber, alloying industry, engraving & photographic process, synthetic rubber, paints, petroleum refining photoconductors, photovoltaic cells, pigments, plastics, and electroplated parts	0.01	Hypertension, lymphocytosis, diabetes mellitus, microcytic hypochromic anaemia, loss of strength and weight, and emphysema	Chen et al. (2018)
5	Zn	Anti-corrosion coating, Zn alloys, can, batteries, paints, chemical industry producing pigments, soldering, and rubber industry	0.8	Diarrhoea, headaches, lethargy, risks of prostate cancer, leukocytes of bronchiolitis, impotence, and depression	Omraei et al. (2011); Okpara et al. (2022)
6	Ni	Ni-based steel, surgical implants, magnets from alnico, electroplating, coinage, catalysts, super-alloys, rechargeable batteries, microphone pods, non-ferrous alloys, dental prostheses, and plumbing fittings plating	0.2	Encephalopathy, anaemia, lung, hepatitis, kidney impairment, diarrhoea, pulmonic fibrosis, gastrointestinal pain, and renal oedema	Zhang and Wang (2015); Hoseinian et al. (2018)

consequences on the vitamin D metabolism, blood, kidneys, blood pressure, and Central Nervous System (CNS) with severe exposure leading to kidney damage, brain damage, and gastrointestinal illness (Parihar et al. 2019). Cd exposure has been linked to persistent kidney tubular damage, an increased risk of gene mutations, decreased semen quality (De Angelis et al. 2017), visual motor impairment, and loss of concentration (Okpara et al. 2022). Ni is the metal most frequently causing concerning human health, and in higher quantities, it could result in cancer and respiratory issues (Mrozik et al. 2021). Symptoms of Zn intoxication include kidney and liver damage, vomiting, icterus, and diarrhoea (Nriagu 2007; Malik et al. 2014).

After discussing the enormous sources of contamination, the focus now shifts to a thorough investigation of the river Yamuna and its conditions in different states. The Yamuna River has distinct features and difficulties as it flows through several Indian states. The next sections will examine each of these segments in detail, providing an overview of the Yamuna River's current condition and the steps that must be taken to ensure its continued existence.

Water quality assessment

Water quality assessment is the process of evaluating the physical, chemical, and biological characteristics of water regarding its natural condition, human influences, and potential applications. Water quality is evaluated to see if it is safe for the environment or acceptable for consumption and other applications, including irrigation and sustenance of aquatic life. This is often done by comparing the physical, chemical, and biological characteristics of the water against a set of specifications (Adelagun et al. 2021). Several physicochemical parameters that are frequently evaluated in research on water quality include water temperature, pH, DO, TDS, turbidity, ammonia, nitrate, HMs, chloride etc. Table 5 gives an overview of the acceptable limits as per different standards.

Table 5 Physico-chemical parameters of some established standards (BIS 10500-2012); CCME (2001); WHO (2011)

Parameter	BIS	CCME	WHO
Water temperature (°C)	–	15	25
pH	6.5–8.5	8.5	6.5–8.5
DO (mg/L)	–	5	–
TDS (mg/L)	500	500	500
Turbidity (NTU)	1–5	5	5
Ammonia (mg/L)	0.5	1.37	0.2
Nitrate (mg/L)	45	48.2	50
Lead (mg/L)	0.01	0.01	0.01
Chromium (mg/L)	0.05	0.05	0.05
Iron (mg/L)	0.3	0.3	0.3
Aluminium (mg/L)	0.03	2.9	–
Calcium (mg/L)	75	–	–
Chloride (mg/L)	250	–	200–300
Copper (mg/L)	0.05	2	2
Fluoride (mg/L)	1	1.5	1.5
Magnesium (mg/L)	30	–	–
Manganese (mg/L)	0.1	0.12	0.4
Sulphate (mg/L)	200	–	–
Total alkalinity (mg/L)	200	–	–
Total hardness (mg/L)	200	–	–
Zinc (mg/L)	5	≤ 5	–
Cadmium (mg/L)	0.003	0.007	0.003
Nickel (mg/L)	0.02	–	0.07

State-wise studies of River Yamuna

The river Yamuna plays a crucial role in supplying drinking water to various regions, including India's capital, Delhi, and neighbouring states such as UK, Haryana, and UP. Fig. 4 shows the map of the different states in which the Yamuna River flows. However, concerns have risen over the last few years because of the decline in the water quality of the river. The discharge of large volumes of moderately treated and untreated effluent into the Yamuna, particularly between Wazirabad and Okhla in Delhi's NCT, has been a significant contributor to this decline. Additionally, the pollution is exacerbated by the waste generated by cities located along the river's bank. Various pollution sources ranging from large factories and urban slums to extensive colonies, affects the health of individuals living in rural areas who consume untreated water from Yamuna River. This collective impact has also caused substantial deprivation of the river's water quality (Sharma and Chaudhry 2015). The Yamuna water diversion, detailed in Table 6 based on information from CPCB in 2000, illustrates the specific locations and constructions where water diversion occurs along the EYC and WYC (CPCB 2000). Table 6 outlines the intended use of diverted water and its subsequent effects on the river's condition downstream. This information provides a concise yet comprehensive overview of the initiatives taken for water management and their effects on the Yamuna River across various states and areas. The barrages, while serving specific purposes, have also created a lotic (flowing) habitat by impeding the river's natural flow (Mishra 2010). Understanding the intricacies of water management and its consequences is essential for addressing the challenges faced by the Yamuna and ensuring sustainable water quality in the region.

Figure 5 shows that according to CPCB (2006), MP has the maximum catchment of river Yamuna while HP has the minimum.

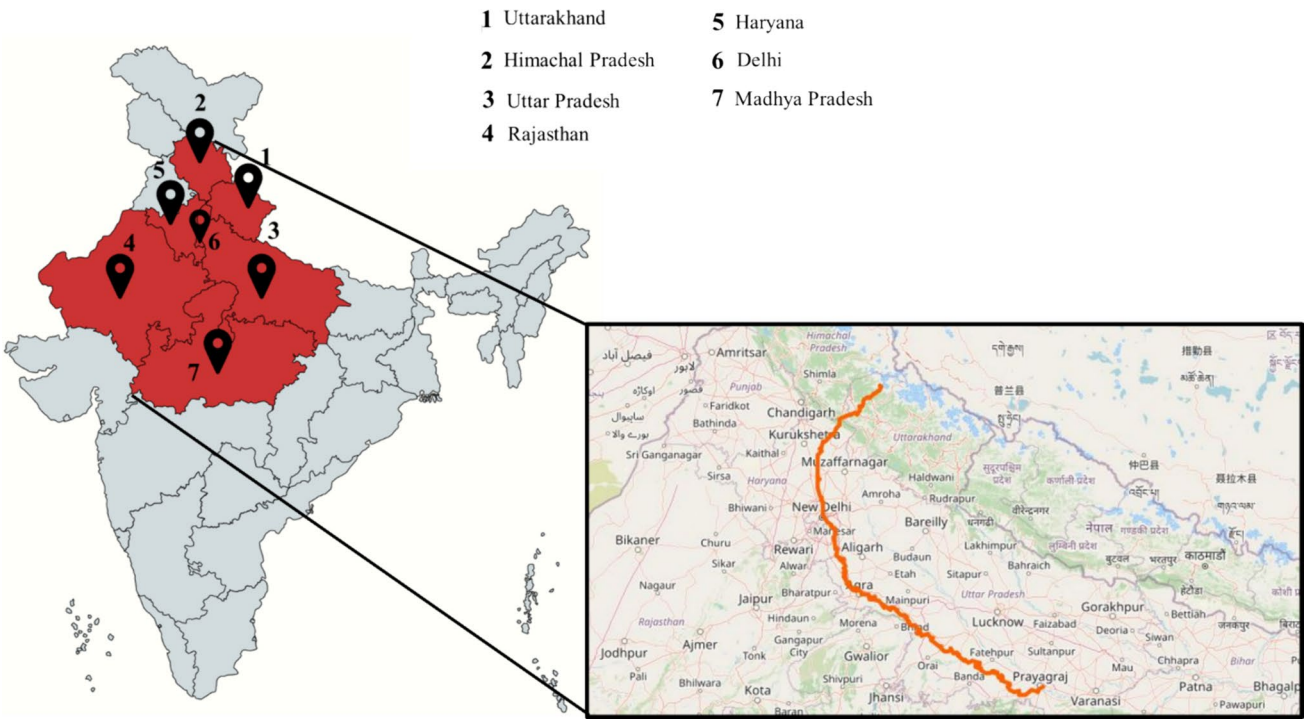

Fig. 4 Map showing the geographical distribution of Yamuna River in India *source* accessed from MapChart and QGIS software

Table 6 EYC and WYC water diversion (CPCB 2000)

Site	Structure	State	Purpose	State of River
Dak Patthar	Barrage	UK	Generation of power	Diverted into canal
Asan	Barrage	UK	Generation of power	Diverted into canal
Hathnikund	Barrage	UP/Haryana	Drinking water and irrigation	Diverted into WYC and EYC (no water flow downstream in dry season)
Wazirabad	Barrage	Delhi	Drinking water	No downstream water flow in dry season
ITO bridge	Barrage	Delhi	Power plant water supply	Available mainly from drains
Okhla	Barrage	Delhi/UP	Agra canal water supply	Absence of water flow downstream in dry season

This section explains a broader perspective of the Yamuna flowing through several states of India. The water quality of the Yamuna River is degrading due to untreated effluent discharge from various sources. The Yamuna Water Diversion, as outlined by the CPCB, emphasizes the need for nuanced water management to ensure sustainable quality.

In the next section a detailed study of individual states i.e. UK, UP, HP, Haryana, Rajasthan, MP, and Delhi will be discussed, understanding the flow of Yamuna through these states. The reason for effluent discharge will be discussed individually for the flowing states.

Uttarakhand

The UK, renowned for its Char Dham pilgrimage sites, including Yamunotri, Gangotri, Kedarnath, and Badrinath, holds significant importance in the confluence of vital river networks (Matta et al. 2012). These locations frequently face environmental challenges exacerbated by increasing water demand, inadequate sewage infrastructure, and a scarcity of wastewater treatment facilities. Yamunotri, positioned near the Yamuna River's mouth, is a site of continuous bathing activities, contributing to the strain on water resources (Semwal and Akolkar 2006). Despite the ecological significance of the upper section of the Yamuna River in the Himalayas, characterized by healthy conditions and minimal sewage discharge, challenges persist due to rising water demand and insufficient wastewater management infrastructure (UePPCB 2022).

A study by Khanna et al. (2010) on the river Yamuna focussed on examining a range of various Physico-chemical parameters along a 20 km stretch of the Yamuna River, from

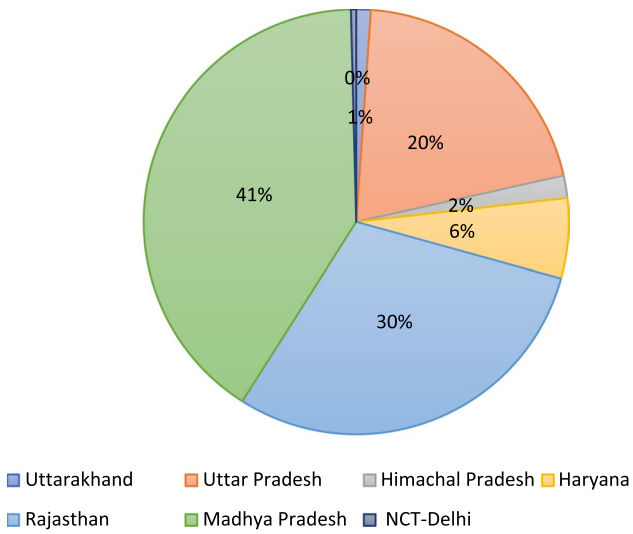

Fig. 5 Percentage state wise catchment of river Yamuna (CPCB 2006)

Kalsi to Dakpatthar, during different seasons. According to the study, of all the different seasons, the monsoon saw the highest level of Total Suspended Solids (TSS), Total Solids (TS), and Total Dissolved Solids (TDS) as well as the temperatures. Furthermore, winter observed the highest temperature and DO. The overall physicochemical parameters were found to be good, but seasonality greatly affected hydrological parameters. An important Yamuna tributary that runs through the UK's Garhwal region, the Tons River is essential to the dynamics of the river. The Tons River water quality in Dehradun, UK, was evaluated by Khanna et al. (2010), who discovered that seasonal variations were seen in pH, temperature, DO, and velocity. Their study also revealed that the primary sources of pollutants in the river were household garbage, agricultural runoff, and industrial discharge. The summertime increased BOD measurements indicated the possibility for bacteria to cause health risks (Khanna et al. 2010). Therefore, Sharma et al. (2015) concluded that industrial post-processing, agricultural residues, and environmental influences are the essential sources of pollution After they analysed the water quality indices of the UK's Yamuna River. Moreover, the principal component analysis proved useful and required in creating relationships linking pollution to the present degree of quality When it comes to tackling big problems, we need to be laser-focussed with our actions. Just take a look at the research on how we humans are interconnected with the delicate ecology of UK's water bodies which is clear that we need sustainable systems to keep our environment safe. From studying the water quality indicators of the Yamuna River in the UK, it was found that the main culprits of pollution were industrial waste, agricultural runoff, and domestic sources. By using Principal Component Analysis (PCA), we were able to draw connections between these pollution sources and the state of our water quality. This highlighted the need for us to take targeted steps to address these issues. The human activities correlate with the poor health of the river ecosystem in the UK just goes to show how crucial it is for us to use sustainable management strategies to protect our local environment.

Himachal Pradesh

The Tons River, which is the Yamuna's biggest tributary, serves as a natural boundary line between the states of UK and HP. (Pandey et al. 2018). Even though it's such a significant part of the landscape, there hasn't been a lot of research done on this particular stretch of the river. Few studies that have been conducted didn't find many problems with the water quality as it stands today.

The pollution issue in HP is getting out of control, especially with all the untreated industrial and household waste being dumped into the Yamuna near Ponta Sahib. It's a harsh situation (Kashyap et al. 2016). In one significant study, Kasyhyap et al. (2016) thoroughly examined physicochemical water quality indicators throughout an 18 km stretch of the Yamuna in the pre- and post-monsoon seasons of 2014 (Kashyap et al. 2016). The study found clear seasonal fluctuations in eight sample areas chosen based on industrial wastewater discharge sites. The post-monsoon season observed the peak of turbidity, but the pre-monsoon witnessed the greatest levels of electrical conductivity (EC), pH, TDS, COD, BOD, Zn, and Fe. Elevated BOD and COD values exceeding permissible drinking water limits indicated significant pollution, attributing the rise to untreated household waste, industrial effluents, and urban runoff from the industrial core. Despite the pollution, the river segment was classified as Class "D," suitable for irrigation, regulated wastewater disposal, wildlife fisheries reproduction, and industrial cooling.

Another study focussed on the Sirmour region of HP, collecting 96 samples of surface water from eight sites along the river Yamuna between 2014 and 2015. The investigation aimed to assess water quality concerning nine HM and three non-metals. Results indicated contributors from both natural as well as anthropogenic activities to the abundance of metals in the river. In 2014 and 2015, identified aspects were held accountable for the degradation of river water quality. There was a noticeably high pollution load in the river lengths throughout 2015, as evidenced by the findings that the metals Cr, As, and Zn were extremely, moderately, and marginally polluting, respectively. Although the study did not show a direct threat to public health, it did highlight the need to halt some downstream operations to restore the region's water quality. The results highlight how closely environmental factors and human activities are connected. It just goes to show how important it is for us to take

sustainable steps to protect HP's precious water resources (Kashyap et al. 2015).

Uttar Pradesh

The Yamuna River, flowing through Agra in U.P., is like a lifeline. It's a vital source of freshwater that furnishes the needs of industries, farms, and households. It's not just a river, it's a provider for the people (Pal et al. 2017). The Yamuna River's watershed is like a complex web, and many drains are crisscrossing the bustling city of Agra (Naushad et al. 2014; Chadetrik et al. 2015). Mathura faces the same problem, with the Yamuna being the main source of fresh water. But the water quality is a big concern because of the densely populated watershed that's all tangled up with numerous drains (Mayank and Tyagi 2013; Pal et. al 2017). The Fort Drain is a major participant when it comes to polluting the river. It's famous for dumping heavy metals into the Yamuna (Biswas et al. 2018). The main factor contributing to the declining water quality in Agra and Mathura is the wastewater from Delhi, transported downstream, which is both untreated or inadequately treated. A study conducted by Issac and Siddiqui (2022), was a thorough, year-long study examining the water quality of nine sources in UP and Agra. Their research dug into the nitty-gritty, measuring 14 different water quality indicators. Apart from a few minor malfunctions like high hardness, TDS, and Sulphate levels during certain times of the year, the water quality was mostly aligned with the standards set by the WHO and the Bureau of Indian Standards. The WQI results indicated seasonal variations in the river water and were safe for drinking during the monsoon, but due to dilution, it was deemed unsafe for consumption in the monsoon, summer, winter or spring. Through PCA, it was possible to pinpoint key elements that significantly influence water quality and elaborate correlations between various quality data.

To identify the variables affecting the primary and secondary irrigation water quality, (Reddy et al. 2021) evaluated 24 water samples in the Chaka block of the Prayagraj district. Their research uncovered that the pH levels were ranging between 6.22 and 7.34, with the average hanging around 6.74. But when it came to the WQI for irrigation, all the samples were in poor condition. The study put the spotlight on the need for extra steps to improve the quality of irrigation water in the area. (Sharma et al. 2022) used satellite Imagine using Landsat satellite images, to look back over a decade (2009–2019) and study the Yamuna River near Balua Ghat, Allahabad. His finding was a bit concerning. Urban areas are consuming the open spaces along the river. After checking out factors like COD, temperature, BOD, and pH, it was clear that the unplanned growth of slums and reckless dumping of waste into the water were affecting the water quality. These findings are of utmost importance, especially as UP strives to meet the growing demands on the Yamuna.

Rajasthan

The Chambal River is a mighty stretch of water that's 960 km long. It starts its journey in Indore, in the state of Madhya Pradesh (MP), and flows its way through Rajasthan, playing a huge part in the region's water body(Sisodiya and Mathur 2021). Acting as a natural border between Rajasthan and MP, it finally meets the Yamuna in U.P. (Kumar et al. 2014). At Pachnada in Bhareh, a place that sits on the border of the UP districts of Etawah and Bhind. The Kwari, Chambal, Yamuna, Pahuj, and Sind rivers all come together in a grand confluence that occurred here (Chouhan et al. 2019).

Abba et al. (2015) conducted a thorough assessment of Chambal River's water quality, using six different criteria. They found that the water quality ranged from 'Fair' to 'Poor'. The water quality of Udi station was rated as 'Marginal to Poor' on the Canadian Council of Ministers of the Environment's grading system, which uses a scale from 0 to 45. They zeroed in on instances from 2000 onwards where the water quality just wasn't cutting it for drinking water standards. The WQI, which uses a scale from 0 to 32, was stamped as 'Poor' during 1999 and 2001–2003 because it didn't meet the cutoff for drinking water standards. The water quality at the station took a spin between 2000 and 2004, often falling way short of what's needed for drinking water.

According to a study published in 2019 by Chouhan et al. (2019), they routinely collected water samples from various locations along the Chambal River, primarily in Kota City. Their investigation revealed concerning trends in several water quality indicators during the pre-monsoon season. Parameters such as COD, EC, TDS, pH, BOD, and DO exhibit signs of depletion and contamination. Remarkably, the surface water quality of the Chambal River was suboptimal during that specific period. Given that the Chambal flows through Rajasthan, these studies underscore the urgent need for coordinated efforts to monitor and mitigate the impacts on this vital river ecosystem.

Haryana

The Yamuna River plays a crucial role in Haryana, flowing through Yamuna Nagar before it reaches Hatnikund. At this point, water is directed into Delhi's drinking water supply and irrigation system via the Western Yamuna Canal (WYC) and Eastern Yamuna Canal. Yamuna Nagar, the second-largest industrial city in Haryana, faces challenges due to high waste generation attributed to population growth, uncontrolled urbanization, and industrial expansion. The river's catchment area up to the Hatnikund headworks spans nearly

12,950 km^2, and during the lean season, inadequate water flow downstream of the Hatnikund barrage leads to the river drying up in various stretches Rai et al. (2012).

Examining the physicochemical features of WYC water at three sampling points, Bhatnagar et al. (2009) found elevated pollutant levels downstream from the influx of sewage and industrial effluents. Parameters such as conductivity, turbidity, alkalinity, free CO_2, hardness, calcium, chloride, magnesium, phosphate, orthophosphate, ammonia, and sulphate registered high levels, while DO showed lower values (DO) (Bhatnagar et al. 2009). DO and BOD emerged as crucial indicators of water quality. The WQI indicated significant pollution at the upper segments of the river, especially before the entry of residential and industrial pollutants. The middle stretch, where a drain carrying domestic waste and industrial effluent joins the river, and 6 km downstream from the point of entry of contaminants, exhibited severe pollution, particularly pronounced during the summer months.

Ravindra et al. (2003) contributed insights into the typical measures of water quality features at various Yamuna canal station locations. Notably, DO, exhibited higher values in winter compared to lower values in summer. Flow variations were evident between monsoon and non-monsoon seasons, with station 1 experiencing larger flows during the monsoon. Changes in water flow were observed in parameters like free CO_2, hardness, alkalinity, and ammonia, decreasing in June and increasing in July, emphasizing the dynamic nature of water quality in the Yamuna River in Haryana (Ravindra et al. 2003). These studies collectively underscore the complex environmental issues faced by the Yamuna as it courses through Haryana, calling for comprehensive strategies to address pollution and sustain the river's health.

Delhi

The Yamuna River, which flows about 855 miles (1376 km) south from the Himalayas through several states, including Delhi, has been significantly affected by pollution. The toxic foam, a mixture of sewage and industrial waste, has formed over sections of the Yamuna River, leading to respiratory and skin problems (Arora and Mehra 2003). Despite its toxicity, many villagers downstream continue to use the water for bathing and even drinking. The river's pollution load is particularly high in areas surrounding Delhi, where the dense population and waste contribute significantly to the contamination (Sehgal et al. 2012). The Yamuna River, which is vital to Delhi but has severe pollution issues because of industrial emissions and urbanization, provides more than 70% of the city's water (Parween et al. 2017). Many drains that discharge significant volumes of pollutants into the river, such as the Shahdara and Najafgarh drains, raise the river's pollution levels (Bhardwaj et al. 2017). Despite its small size, the watershed region has substantial challenges with development and pollution. The Yamuna's pollution load is increased by the heavily industrialized districts' frequent waste releases, particularly during the dry seasons (Verma et al. 2022).

Research conducted by Madan et al. (2018) focussed on the Yamuna River in Delhi and the WQI of an industrial outlet. In their research, specifically studied the Yamuna River in Delhi and assessed the WQI of an industrial outlet. Their findings point to the Nizamuddin outflow point, the DO level was alarmingly low, registering 0% DO. This indicates extreme pollution in that section of the river and the gas turbine station and thermal power plant outputs also exhibited significant WQI values. These findings underscore the severe pollution in these industrial areas. Another study by (Nehra and Singh 2020), in their research, specifically studied the Yamuna River in Delhi with a particular focus on the Najafgarh drain. He claims that the Yamuna in Delhi has essentially transformed into a sewer. Without proper treatment, it is unusable due to severe contamination and The Najafgarh Drain carries significant volumes of organic waste and a high BOD load. This pollution could potentially pose a hazard to both the food chain and human health

Kaur et al. (2021) used ciliate populations as bio-indicators to evaluate how the Yamuna River's water quality affected the microbiological life. The Yamuna River's ciliates have been proven to be reliable bioindicators of pollution, and the study found that the river's quality significantly declined as it passed through Delhi NCR (Kaur et al. 2021 Table 7 of (CPCB (2021) Annexure VI (a)) shows that Delhi's water quality, like that of other states including Haryana and UP, struggles to meet the standards required for bathing. These findings underscore the pressing need for all-encompassing actions to mitigate pollution issues and protect the health of Delhi's Yamuna.

Madhya Pradesh

MP also makes the vast network of the Yamuna River even richer, while the river’s tributaries present the essential signs of the general health of the system. The most prominent tributaries are underlined below. The Betwa River implies the main stream of the Betwa River Basin. A scholarly analysis of the periodic fluctuation of the physicochemical properties of the Betwa River was carried out by (Vishwakarma et al. 2013) between October 2011 and Aug 2012. Substantially, high COD, BOD, and hardness were noticed in Nayapura and Mandideep, which can be interpreted as evidence of pollution in these sites. The present quality of the river of Betwa was characterized by low pH and DO in Vidisha, Bhojpur, and Jhirri, demonstrating the dynamics of the river’s water quality.

The physicochemical properties of the Ken River, another Yamuna tributary in MP, were studied by Awasthi

Table 7 Water quality data of River Yamuna (CPCB 2020) (CPCB 2021 annexure VI (a))

S. no	State	Location	Temperature (°C)	pH	DO (mg/L)	Conductivity (μmhos/cm)	BOD (mg/L)	FC (MPN/100 mL)	TC (MPN/100 mL)
1	UK	Yamunotri*	–	7.4	10.8	79	BDL	4	4
2	HP	Hanumanchatti*	–	7.9	11.6	145	BDL	94	140
		U/S Lakhwar Dam	16–22	7.7–8.4	8.6–10.8	142–320	0.8–1.0	2–9	2–30
		U/S Dak Patthar	17–24	8.0–8.2	8.2–10.2	106–162	1.0–1.2	17–34	40–70
		U/S Paonta Sahib	13–28	6.4–8.0	7.1–9.1	214–5866	0.4–1.8	9–46	32–140
		D/S Paonta Sahib	11–31	6.6–7.7	7.0–9.0	248–448	0.4–2.0	11–27	38–94
		U/S Ranbaxy	16–34	7.0–7.6	7.1–8.9	162–605	0.4–1.6	10–13	32–94
		D/S Ranbaxy	16–34	7.0–7.6	7.0–8.8	161–2593	0.4–1.8	10–33	33–110
3	Haryana	Hathnikund, Yamunanagar	21–29	7.3–8.3	7.1–8.5	192–620	2.4–6.6	200–3200	1100–21200
		Kalanaur, Yamuna Nagar	19–29	7.2–8.3	6.8–9.3	197–608	2.2–5.2	200–49000	3900–1600000
		Magalaura	–	7.2–8.5	5.2–10.8	157–451	BDL-4.0	49–7000	230–35000
		Khojkipur Panipat	21–29	6.6–8.4	5.0–8.4	255–942	1.4–7.0	90–70000	270–278000
		Sonepat	14–35	6.5–8.6	7.1–10.6	235–683	1.8–5.5	110–220000	330–1600000
4	Delhi	Palla	16–33	7.6–9.0	5.6–17.1	273–789	1.4–7.9	78–79,000	330–350000
		Nizamuddin	13–31	7.1–7.9	BDL-2.4	460–1641	5.6–57.0	210,000–11000000	700,000–28000000
		Okhla Bridge (Inlet of Agra Canal)	15–31	7.1–7.9	BDL-2.6	488–1526	5.6–27.0	78,000–3300000	350,000–16000000
		Okhla After Meeting of Shahdara Drain	15–33	7.2–7.8	BDL-5.6	499–1773	7.7–114.0	68,000–7900000	490,000–35000000
		Agra Canal at Madanpur Khadar	15–31	7.1–7.8	BDL-1.4	573–1541	11.4–32	330,000–4900000	1,100,000–24000000
5	UP	Mazawali	14	6.4–8.1	BDL-2.3	978–1719	8.7–59.0	680–4900000	3300–9200000
		Shahpur	14–32	7.1–7.5	4.6–6.9	–	5.8–10.0	31,000–85000	53,000–110000
		Kesighat, Vrindavan	15–32	7.1–7.6	4.0–7.6	–	6.0–10.4	24,000–65000	46,000–88000
		Mathura U/S	13	6.5–8.5	BDL-4.6	1096–1816	7.3–30.0	400–920000	1100–9200000
		Vishramghat, Mathura	14–32	6.9–7.6	4.1–7.4	–	6.8–12.0	31,000–92000	54,000–140000
		Mathura D/S	12	6.5–8.3	1.6–6.8	1195–1903	6.9–21.0	1700–170000	2200–1400000
		Agra U/S	12	6.5–8.5	1.4–9.6	1154–1872	4.7–25.0	220–3300000	2200–7900000
		D/S of Agra	–	6.5–8.7	3.8–16.2	875–1920	9.1–25.0	450–6100000	2200–32000000
		Bateswar	–	7.8–7.9	2.6–9.2	1054–1471	14.0–20.0	450–1400	3300–13000
		Etawah	–	7.7–7.9	7.3–7.3	1288–1507	13.0–15.0	20–200	1300–1700
		Juhika B/C with Chambal, Etawah	–	7.7–8.2	8.1–11.3	846–937	8.7–13.0	68–7800	1700–220000
		Hamirpur	–	8.0–8.4	0.4–8.4	750–832	2.8–6.0	330–2200	1300–13000
		Allahabad D/S (Balua Ghat)	22–27	7.8–8.4	7.3–11.5	418–725	1.8–2.4	310–2100	1700–3900

U/S—up stream; D/S—down stream

*Locations monitored on yearly basis. BDL value for DO is 3.0 mg/L

et al. (2018). Numerous data were analysed at seven monitoring sites in and around the Panna District, including pH, temperature, EC, turbidity, TDS, TSS, DO, COD, BOD, chloride, phosphate, sulphate, nitrate, alkalinity, total hardness, calcium and magnesium hardness. The conditions were very alkaline, as shown by the pH values, which varied from 7.21 to 7.22. Average conductivity varied seasonally from 600 to 1100 µmhos/cm in post-monsoon and from 900 to 1200 µmhos/cm in pre-monsoon and monsoon seasons. The acceptable levels of DO, COD, and total coliform that were maintained in the river are proof that they correspond with WHO drinking water requirements. However, the study highlighted that continuous monitoring of the Ken River in the MP region is necessary to ensure its well-being (Awasthi et al. 2018).

Reflecting the geography of the Malwa Plain, the Chambal River is the main tributary of the Yamuna that flows through central MP (Khan 2022). At Janapao temple, about 24 km southwest of Mhow in MP, the river's origins in the spectacular southern slopes of the Vindhya Mountains are reflected in its veins (Hussain et al. 2011). In addition to the river's original upstream flow, three smaller streams, or nallahs, located next to the temple improve the river's beauty and biodiversity (Saksena et al. 2008). Besides its pristine reputation, the river has a wide variety of aquatic species. These species include eight freshwater species such as turtles, skimmers, Gangetic River dolphins, and two species of crocodiles (gharial and mugger; Saksena et al. 2008).

Sohan thoroughly examined the chemical and physical properties of the Chambal River in the National Chambal Sanctuary in MP (Yadav et al. 2014). This sanctuary stretches over 600 km downstream from Kota to Yamuna confluence and is a buffer zone. The Chambal River in this sanctuary has been classified as Class C by the CPCB based on extensive data collected during 2003 and 2012. This classification indicates that the river in the sanctuary region is pollution-free and homely for a variety of aquatic plants and animals, including some endangered species. The findings highlight the success of conservation efforts in maintaining the ecological integrity and health of the Chambal River in this critical area (Yadav et al. 2014). These studies shed light on many of the dynamics of water quality in MP rivers and highlight the intricate interrelationships among various natural processes that sustain the Yamuna.

This section explains how the Chambal River being a tributary to Yamuna flows through MP. This river is free from pollution and is home to a variety of aquatic species. Several tests have been done to prove how good this river is. The next section deeply explains the observation of water concerning the period of COVID-19. Urbanization degraded Yamuna River's water, but COVID-19 lockdowns unexpectedly improved it.

Observations of water quality with respect to period of COVID-19

Scientists globally have extensively studied the quality of the Yamuna River water, recognizing a concerning decline attributed to industrialization, urbanization, and modifying land use patterns over recent decades (Thakur et al. 2018; Dudeja et al. 2011). However, an unexpected turn of events occurred with the outburst of the Coronavirus Disease (COVID-19) in late 2019, causing widespread disruptions (Awasthi et al. 2021). This global health crisis inadvertently led to positive changes in the water quality of the Yamuna River, particularly during and after lockdowns implemented in India. (Arora et al. 2020).

In the wake of the COVID-19 epidemic, lockdown measures brought industrial activities to a dead end, resulting in a remarkable improvement in the Yamuna River's surface water qualityThe Yamuna section of Delhi's WQI drastically improved, according to Patel et al. (2020), with notable decreases in COD and BOD of 39.3% and 42.8, respectively, and a 40% decrease in faecal coliform (Patel et al. 2020). Other research, such as the one on the Yamuna tributary Assan River by Arora et al. (2020), confirmed similar positive trends. Improvements in pH, TDS, and EC were seen at many test locations in the research (Arora et al. 2020). Arif et al. (2020) also observed notable decreases in the EC, pH, BOD, DO, and COD levels in Delhi's Yamuna River during the shutdown (Arif et al. 2020).

Singh et al. (2020) addressed how crucial Yamuna River water quality parameters were impacted by economic decisions. Their findings verified that significant increases in water quality measures including TSS, BOD, and COD were positively connected with lockdown-induced activity elimination (Singh et al. 2020). Studies done after 2019 consistently show the impaired status of the Yamuna River's water quality, which is further supported by the data summary in Table 8 and the observed positive trend. However, it is crucial to note that even during the lockdown period, challenges persisted, with untreated sewage flow identified as a significant concern (Patel et al. 2020; Ahmed et al. 2021). This emphasizes the ongoing need for effective wastewater management and domestic sewage treatment to sustainably address water pollution.

To fulfil the primary goal of the present review paper, i.e. to discuss the different investigations that assess the water quality of the Yamuna, a thorough search was taken forward using the electronic search engine "ScienceDirect" by using the keyword "Yamuna" and organized in Table 8. Only peer-reviewed research publications written in English and published between 2019 and 2023 (after the first lockdown of the COVID-19 pandemic) were chosen

Table 8 Various research studies on Yamuna River from 2019to 2023

References	Location	Objective	Result
Thakur et al. (2023)	Alluvial region of the Yamuna basin, India	To evaluate the groundwater quality in the Yamuna basin's alluvial zone, which mostly consists of unconsolidated sand, silt, and clays	The WQI ranges from excellent to inappropriate groundwater is harmed by fertilizer and fungicide leaching
Mishra et al. (2023)	Yamuna River, Delhi	Evaluation of the amount of organic micropollutants in the perennial Yamuna	47 organic micropollutants during non-monsoon; 55 during monsoon. In the monsoon, organic micropollutant concentrations increased by almost 50%
Khan et al. (2023)	Yamuna river basin, Northern India	To evaluate how human-induced activities affect groundwater quality	WQI: poor water (34.2% pre-, 52.63% post-monsoon). HCO_3 and SO_4 cause excess hardness. Groundwater mostly suitable for agriculture
Ahmed et al. (2022)	Yamuna River, northern India	To evaluate groundwater contamination while highlighting hotspots for HMs	Groundwater samples classified: very poor (77.3%), poor (13.6%), good (9.10%). Escalation of waste and agriculture impacts aquifers
Sankhla et al. (2022)	Yamuna River, Delhi, India	To assess the effect of variations in climatic conditions on Pb and Ni concentrations	Pb and Ni exceeded WHO limits. Concentrations rise with higher temperature and lower humidity
Parween et al. (2021)	River Yamuna, Delhi	To estimate the health risks related with persistent insecticides and HMs, as well as their toxicity	Pollution load highest at Yamuna exit in Delhi
Kumar et al. (2021)	River Yamuna	Yamuna River Cr Bioremediation with Cyanobacteria under White and Red Illumination	Spirulina sp. Excels in scavenging Cr (VI) and Cr (III) ions. No harmful environmental effects observed
Biswas and Vellanki (2021)	River Yamuna, India	To find 16 emerging contaminants in the Yamuna River, such as drugs, cosmetics, and hormones	Emerging contaminants found even in sparsely populated up-stream Himalayan area. Emerging contaminants detected in finished drinking water, suggesting treatment inefficiency
Khan et al. (2021)	Yamuna River, Delhi	To investigate the seasonal change of high-priority pharmaceutical residues	Highest predicted no-effect concentration exceedances: ibuprofen (92%), carbamazepine (69%), simvastatin (64%)
Ahmed et al. (2021)	Yamuna river basin, Mathura, UP	To assess the shallow water aquifer's scaling potential and corrosive behaviour utilising geospatial methods and corrosion indices	More than 90% of samples have a chloride sulphate mass ratio > 0.5, indicating corrosiveness towards distribution systems
Saini et al. (2020)	EYC, Saharanpur, UP	To assess hydrokinetic energy	To achieve greatest flow velocity, 904 arrays with 2 units each are needed to harness the hydro-kinetic potential of the Yamuna canal
Khan et al. (2020)	Yamuna River basin, India	To identify artificial groundwater recharge sites based on RS and GIS	26% of the study area has poor groundwater conditions (32–42 m bgl)
Khan et al. (2020)	Yamuna River, New Delhi, India	To investigate the usage of acclimatized microalgae as an acclimatizing medium for the elimination of HMs	High HMs in Yamuna River inhibit growth. A. platensis potential for HM sequestration and pigment production demonstrated
Khan et al. (2020)	River Yamuna, Delhi	Research of the metagenome that focuses on genes for antibiotic resistance found in sediments	Found high levels of macrolide, aminoglycoside, tetracycline, and beta-lactam resistance genes
Patel et al. (2020)	Yamuna River, Delhi	To examine water quality during COVID-19 lockdown period	Turbidity and suspended particulate matter decrease in some areas. Domestic sewage significantly increases pollution despite the absence of industrial effluents

Table 8 (continued)

References	Location	Objective	Result
Kaur et al. (2020)	Yamuna sub-basin, Panipat	Groundwater potential evaluation of an alluvial aquifer using RS and GIS methods in unification with catastrophe theory and analytical hierarchy method	Poor groundwater potential assessment due to human actions
Lamba et al. (2020)	River Yamuna, Delhi, India	Sewage mediated transfer of antibiotic resistance	Poor connectivity between sewage treatment plants and drains leads to sewage emission, increasing river pollution. Untreated sewage release into the Yamuna River poses a high health risk, emphasizing the urgency of comprehensive sewage treatment
Kaur et al. (2019)	River Yamuna, Panipat, Haryana	Hydrogeochemical characterization of groundwater	Groundwater analysis doesn't show supremacy of a single rock mineral dissolution route
Samson et al. (2019)	Yamuna River, River Ganges, India	Metagenomic study of microbial communities in confluence stretch of Yamuna and Ganges Rivers	Functional profiles at the confluence were influenced by the Yamuna, not taxonomic diversity. Profiles were similar pre-confluence (Ganges) and farther downstream of the confluence
Ahamad et al. (2019)	Yamuna River, New Delhi, India	Impact of non-engineered Bhalswa landfill on groundwater from Quaternary alluvium and potential human health risk	Groundwater samples show elevated electrical conductivity and major cations/anions. Most HM concentrations exceed WHO limits except for Cr and Cu

for the present investigation by considering the keyword "Yamuna" as "title only" from the advanced search option of ScienceDirect's search platform. Initially, 67 articles were extracted, which were reduced to 25 articles by choosing the years "2019–2023" and the article type as "research articles". Four articles were eliminated manually because of their irrelevancy concerning the goal of the review, and finally, 21 articles are summarized bibliographically in Table 8.

In Table 8, recent studies, spanning 2019–2023, consistently portray a grim picture of the Yamuna water quality. The studies mentioned in the table indicate a related pattern of degradation of water quality in the Yamuna. Several contaminants, encompassing HMs (Ahmed et al. 2022), and organic micropollutants (Mishra et al. 2023) cause serious risks to both aquatic ecosystem and human health (Ahamad et al. 2019). Pharmaceutical remains, which are the emerging contaminants, were detected even in the lightly inhabited upstream Himalayan areas (Biswas and Vellanki 2021). Results indicate that pollution problems affect not just heavily populated metropolitan areas but also the purer river regions. Ahmed et al. (2021) found that untreated sewage flow into the Yamuna River is still a serious issue (Ahmed et al. 2021). Domestic sewage persisted in increasing water pollution, even with the absence of industrial effluents during the COVID-19 lockdown period (Patel et al. 2020) which emphasizes the requirement of management sources and thorough domestic sewage treatment. One of the investigations used Remote Sensing (RS) and the Geographical Information System (GIS) for the identification of the artificial groundwater recharge site that helps in balanced water resource management (Khan et al. 2020). Overall, Table 8 highlights the increasing water quality disputes involving organic microplastics and HMs, emphasizing the critical need for conservation and water pollution control strategies to protect the Yamuna River water resource and its ecosystem. In the coming section, different management, and treatment strategies to control and purify the wastewater will be discussed.

Management and treatment of wastewater

The Yamuna River, integral to the lifeline of cities such as Mathura, Delhi, Agra, and Etawah, poses severe pollution challenges resulting from rampant industrialization and urbanization, notably in key areas like Mathura, Delhi, and Agra (Mishra 2010). To address this crisis, a bottom-up approach is deemed more effective than a top-down one. Rehabilitation efforts demand a multifaceted strategy, encompassing education, heightened public awareness, and improved watershed management. Key actions critical to enhancing Yamuna's water quality include bolstering

environmental management, refining agricultural practices, optimizing solid waste management, fostering water conservation, improving water use efficiency, allocating funds for wastewater management, promoting wastewater reuse and recycling, advocating wastewater treatment and technology, upgrading sewage treatment plants, and ensuring proper sewage disposal.

The global availability of clean water, essential for ecosystem development and socioeconomic sustenance, faces rapid depletion (Wu et al. 2019). Wastewater, stemming from various sectors, like households, industries, and municipalities, is a significant contributor to freshwater challenges (Hafeez et al. 2021). A staggering 56% of the 3928 km^3 of freshwater withdrawn annually worldwide is released as untreated wastewater, underscoring the severity of the issue (Morali et al. 2016).

Wastewater treatment is pivotal for sustaining renewable water resources and fostering financial growth at the state level (Shehzad et al. 2020). Unfortunately, this truth often goes underappreciated and underutilized. Almost 2/3rd of the worldwide population grapples with severe water shortages for at least one month each year, exacerbated by a lack of incentives for industries to treat wastewater and ineffective enforcement of pollution control rules by local governments (Hafeez et al. 2021). An integrated strategy, incorporating affordable and sustainable technologies, is imperative to bolster the inclination towards effective wastewater treatment (Hassan et al. 2020).

Wastewater discharges into rivers not only exacerbate water shortages but also contribute to waterborne infections and adversely impact ecosystems and marine life. The management and treatment of wastewater play a crucial in mitigating these challenges by reducing or eliminating sediments, organic matter, disease-causing organisms, nutrients, and other contaminants before the treated water is released back into river ecosystems (Mishra 2010). Effective wastewater management is, therefore, an indispensable component of safeguarding both water resources and the health of river ecosystems. The upcoming section addresses the limitations of this review, serving as potential research gaps for future studies.

Limitation

Although this review study offers insightful information about the River Yamuna's water quality, certain inherent limitations should be noted. Data on water quality may be limited over the complete stretch of the River Yamuna, with certain areas having more comprehensive data than others. Due to the numerous and frequently related pollution sources present along the Yamuna River, pinpointing the exact root cause of pollution can be difficult. Although the review discusses the possible effects of HMs on human health, more scientific studies are necessary for a more thorough health impact evaluation. The amount of thorough research on the water quality of the river Yamuna may be constrained regardless of the inclusion of studies from many states, especially in some areas and some regions that still need to be studied. Although the study emphasizes the significance of wastewater management, it is still necessary to consider the challenges of establishing and maintaining effective wastewater treatment systems across various geographic locations.

Conclusion

The Yamuna River's dire situation necessitates swift action and strict regulation to save the millions of people who depend on its waters as well as the river's natural health. The intricacy of the problem at hand is highlighted by the variety of pollution sources, ranging from untreated home sewage to industrial discharge and agricultural runoff. However, comprehensive policies that systematically target important areas—this report primarily focuses on industrial regulations, wastewater management, sustainable agriculture practices, and campaigns—also provide optimism. These key concepts offer an integrated approach to addressing the river's numerous issues and are crucial cornerstones for restoring the Yamuna River. The management strategies will not only help decision-makers to restore the water quality of the river Yamuna but also all the rivers.

Wastewater management is critical in developing efficient sewage treatment plants (STPs) and decentralized wastewater solutions. This plan attempts to deal with the widespread untreated wastewater outflow, which accounts for a significant 80% of surface water pollution. Cities like Delhi are particularly affected by this problem. Enforcing strong industrial limits is essential to reduce pollution at the same time. When coupled with incentives for greener production methods, these policies should reduce the number of HMs like Cr and other pollutants from unregulated enterprises like Wazirpur. By ensuring that companies adhere to environmentally friendly practices, the impact on the Yamuna's water quality may be reduced. To reduce the high nitrate and phosphate concentrations that lead to eutrophication and fish mortality in the river, it is essential to promote sustainable agricultural practices, mainly through the adoption of organic farming techniques. It is possible to significantly reduce the quantity of hazardous nutrients that enter the Yamuna by using fewer chemical fertilizers and switching to ecologically friendly farming methods. Programs that raise public awareness are essential for simultaneously encouraging a sense of accountability and active participation. Through cleanup initiatives and reporting mechanisms for identifying and penalizing offenders, these programs aim

to create a common commitment to maintaining the river's health. The sense of accountability and ownership generated by citizen engagement ultimately determines how successful more considerable environmental conservation efforts are.

Recognizing the significance of the four primary concerns outlined above, it is evident that the improvement in Yamuna water quality during the COVID-19 lockdowns acts as a sore reminder of what happens when policy interventions are carried out systematically. However, these initiatives must cover more land than simply major cities; they must also address knowledge gaps and sources of contamination in the river's upper reaches, including states like the UK and HP. Preserving the ecological integrity of the Yamuna requires concerted action across political boundaries and socioeconomic groups. This collaborative effort is built upon a foundation of comprehensive scientific research, well-informed policy based on local realities, and grassroots behaviour change mobilization. The restoration of the Yamuna essentially requires a paradigm shift in our relationship with this significant river. Even though it is frequently seen as merely a trash conduit, it plays a crucial role in a variety of ecosystems. To protect the environment for the benefit of present and future generations, people must work together as stewards of these historic rivers to restore their purity.

Future perspective

The Yamuna River is regarded as the spiritual lifeline for millions of Indians. It does, however, currently face an environmental catastrophe that calls for immediate action as well as substantial regeneration initiatives. Subsequent research is sparked by recent foundational studies that pinpoint important knowledge gaps and rank areas that require urgent attention. Understanding the different causes of pollution, such as untreated sewage and industrial effluents, is crucial to developing targeted and successful regulations and cleaning initiatives. Technological developments, especially in the field of artificial intelligence, offer novel instruments for analysing pollutants in rivers such as the Yamuna. Machine learning algorithms are trained on large datasets to help them detect new threats and anticipate when action is most likely to be needed.

The Yamuna's surrounding industrial activities and constant urbanization make longitudinal research essential. These studies provide insightful information on the temporal patterns that affect the river's health over extended periods. The ability of contemporary sensors to identify pollution instantaneously gives real-time solutions and has a lot of potential. By acting as early warning systems that alert authorities to potential threats, these innovations have the potential to drastically alter the way water quality is managed. A paradigm shift towards environmental stewardship is demonstrated by preventive actions such as these, which prioritize prevention over cure—a crucial stance for the long-term, sustainable preservation of natural resources.

This study focuses on directing efforts to restore an ecosystem that was previously flourishing but is now burdened with obstacles caused by people. It also actively establishes strategic paths for future research, pointing out knowledge gaps. Worldwide efforts for environmental conservation, which emphasize the value of safeguarding natural resources like the Yamuna River for future generations as well as the feelings of those who regard the river as sacred, are in harmony with our initiative.

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Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Ethical Approval This article is a review of previously published studies. It does not contain any new studieswith human participants or animals performed by any of the authors. Therefore, no ethical approval is required.

Informed Consent Not applicable as this article does not involve any new studies with human participants.

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References

Abba SI, Sunusi AM, Yakasai YS (2015) Appraisal of yearly and seasonal water quality variation at Chambal river using Canadian Council of Ministry of Environment Moded (CCME). Int J Adv Res Sci Eng 4(01):807–816

Adelagun ROA, Etim EE, Godwin OE (2021) Application of water quality index for the assessment of water from different sources in Nigeria. Promising techniques for wastewater treatment and water quality assessment. IntechOpen, Houston

Ahamad A, Raju NJ, Madhav S, Gossel W, Wycisk P (2019) Impact of non-engineered Bhalswa landfill on groundwater from quaternary alluvium in Yamuna flood plain and potential human health risk, New Delhi, India. Quat Int 507:352–369. https://doi.org/10.1016/j.quaint.2018.06.011

Ahmed S, Sultan MW, Alam M, Hussain A, Qureshi F, Khurshid S (2021) Evaluation of corrosive behaviour and scaling potential of shallow water aquifer using corrosion indices and geospatial approaches in regions of the Yamuna river basin. J King Saud Univ Sci 33:101237. https://doi.org/10.1016/j.jksus.2020.101237

Ahmed S, Akhtar N, Rahman A, Mondal NC, Khurshid S, Sarah S, Khan MMA, Kamboj V (2022) Evaluating groundwater pollution with emphasizing heavy metal hotspots in an urbanized alluvium watershed of Yamuna River, northern India. Environ Nanotechnol, Monit Manag 18:100744. https://doi.org/10.1016/j.enmm.2022.100744

Akhtar N, Syakir Ishak MI, Bhawani SA, Umar K (2021) Various natural and anthropogenic factors responsible for water quality degradation: a review. Water 13(19):2660. https://doi.org/10.3390/w13192660

Anderson EP, Jackson S, Tharme RE, Douglas M, Flotemersch JE, Zwarteveen M, Lokgariwar C, Montoya M, Wali A, Tipa GT, Jardine TD, Olden JD, Cheng L, Conallin J, Cosens B, Dickens C, Garrick D, Groenfeldt D, Kabogo J, Roux DJ, Ruhi A, Arthington AH (2019) Understanding rivers and their social relations: A critical step to advance environmental water management. Wires Water 6(6):e1381. https://doi.org/10.1002/wat2.1381

Arif M, Kumar R, Parveen S (2020) Reduction in water pollution in Yamuna River due to lockdown under COVID-19 pandemic. ChemRxiv. https://doi.org/10.26434/chemrxiv.12440525.v1

Arora J, Mehra NK (2003) Species diversity of planktonic and epiphytic rotifers in the backwaters of the Delhi segment of the Yamuna River, with remarks on new records from India. Zoolog Stud 42(2):239–247

Arora S, Bhaukhandi KD, Mishra PK (2020) Coronavirus lockdown helped the environment to bounce back. Sci Total Environ 742:140573. https://doi.org/10.1016/j.scitotenv.2020.140573

Arora G, Sharma T, Taijas KK, Pant P, Gupta C, Sharma RK (2023) Rejuvenation and restoration of surface water quality amid COVID-19 lockdown: a comprehensive review in Indian context. Environ Eng Res. https://doi.org/10.4491/eer.2022.144

Arun L, Prakash DR, Rout C (2015) Assessment of water quality of the Yamuna River in rural and semi-urban settings of Agra. India Int J Earth Sci Eng 8(4):1661–1666

Awasthi A, Tripathi SK, Tiwari AK (2018) Physico-chemical analysis of Ken river water in Panna district Madhya Pradesh, India. Res J Sci Tech 10(2):131–136. https://doi.org/10.5958/2349-2988.2018.00019.0

Awasthi A, Sharma A, Kaur P, Gugamsetty B, Kumar A (2021) Statistical interpretation of environmental influencing parameters on COVID-19 during the lockdown in Delhi. India Environ Dev Sustain 23(6):8147–8160. https://doi.org/10.1007/s10668-020-01000-9

Awasthi A, Dewan R, Singh G (2022a) Glacier melting: drastic future. Adv Behav Based Saf: Proc HSFEA 2020:255–266. https://doi.org/10.1007/978-981-16-8270-4_19

Awasthi A, Vishwakarma K, Pattnayak KC (2022b) Retrospection of heatwave and heat index. Theor Appl Climatol 147:589–604. https://doi.org/10.1007/s00704-021-03854-z

Bhardwaj R, Gupta A, Garg JK (2017) Evaluation of heavy metal contamination using environmetrics and indexing approach for River Yamuna, Delhi stretch, India. Water Sci 31:52–66. https://doi.org/10.1016/j.wsj.2017.02.002

Bhateria R, Jain D (2016) Water quality assessment of lake water: a review. Sustain Water Res Manag 2:161–173. https://doi.org/10.1007/s40899-015-0014-7

Bhatnagar A, Chopra G, Malhotra P (2009) Water quality indices and abiotic characteristics of western Yamuna canal in Yamunanagar, Haryana. J Appl Nat Sci 1(2):149–154. https://doi.org/10.31018/jans.v1i2.55

BIS (2012) Drinking water specification, Bureau of Indian Standards. New Delhi, 10500. http://bis.org.in/bis/html/10500.html

Biswas P, Vellanki BP (2021) Occurrence of emerging contaminants in highly anthropogenically influenced river Yamuna in India. Sci Total Environ 782:146741. https://doi.org/10.1016/j.scitotenv.2021.146741

Biswas B, Jain S, Rawat S (2018) Spatio-temporal analysis of groundwater levels and projection of future trend of Agra city, Uttar Pradesh, India. Arab J Geosci 11:1–18. https://doi.org/10.1007/s12517-018-3577-4

Cao W, Wang Z, Ao H, Yuan B (2018) Removal of Cr(VI) by corn stalk based anion exchanger: the extent and rate of Cr(VI) reduction as side reaction. Colloids Surf A Physicochem Eng Asp 539:424–432. https://doi.org/10.1016/j.colsurfa.2017.12.049

CCME (2001) Canadian Council of Ministers of Environment: Water Quality Index User's Manual. Canadian Water Quality Guidelines for the Protection of Aquatic Life. pp 1–5

Central Pollution Control Board, "Water Quality status of Yamuna River," New Delhi, April 2000. https://cpcb.nic.in/

Chadetrik R, Arun L, Prakash DR (2015) Assessment of physico-chemical parameters of River Yamuna at Agra region of Uttar Pradesh, India. Int Res J Environment Sci 4(9):25–32

Chen YY, Yu SH, Jiang HF, Yao QZ, Fu SQ, Zhou GT (2018) Performance and mechanism of simultaneous removal of Cd(II) and Congo red from aqueous solution by hierarchical vaterite spherulites. Appl Surf Sci 444:224–234. https://doi.org/10.1016/j.apsusc.2018.03.081

Chouhan RK, Bansal AK, Chhipa RC (2019) Physico-chemical analysis of water at selected point in Kota, Rajasthan. J Drug Deliv Ther 9(4-A):722–725. https://doi.org/10.22270/jddt.v9i4-A.3557

CPCB (2006) WATER QUALITY STATUS OF YAMUNA RIVER (1999 – 2005) Acessed on 4 July, 2024. https://yamunariverproject.wp.tulane.edu/wp-content/uploads/sites/507/2021/01/cpcb_2006-water-quality-status.pdf

CPCB (2020) ANNUAL REPORT 2020-21, Accessed on 4th July, 2024. https://cpcb.nic.in/openpdffile.php?id=UmVwb3J0RmlsZXMvMTQwM18xNjU1MzU0NzkxX21lZGlhcGhvdG8xNjQ3MS5wZGY=

CPCB (2021) ANNUAL REPORT 2020-21 Accessed on 4th July, 2024. https://cpcb.nic.in/openpdffile.php?id=UmVwb3J0RmlsZXMvMTQwM18xNjU1MzU0NzkxX21lZGlhcGhvdG8xNjQ3MS5wZGY=

Dar JA (2023) Analysis of the water quality status of the Yamuna river in the Delhi region

De Angelis C, Galdiero M, Pivonello C, Salzano C, Gianfrilli D, Piscitelli P, Lenzi A, Colao A, Pivonello R (2017) The environment and male reproduction: the effect of cadmium exposure on reproductive function and its implication in fertility. Reprod Toxicol 73:105–127. https://doi.org/10.1016/j.reprotox.2017.07.021

Dinari M, Haghighi A (2018) Ultrasound-assisted synthesis of nanocomposites based on aromatic polyamide and modified ZnO nanoparticle for removal of toxic Cr(VI) from water. Ultrason Sonochem 41:75–84. https://doi.org/10.1016/j.ultsonch.2017.09.023

Dudeja D, Bartarya SK, Biyani AK (2011) Hydrochemical and water quality assessment of groundwater in Doon Valley of outer Himalaya, Uttarakhand, India. Environ Monit Assess 181:183–204. https://doi.org/10.1007/s10661-010-1823-7

Fiyadh SS, Alardhi SM, Omar MA, Aljumaily MM, Saadi MAA, Fayaed SS, Ahmed SN, Salman AD, Abdalsalm AH, Jabbar NM, Shafi AE (2023) A comprehensive review on modelling the adsorption process for heavy metal removal from wate water using artificial neural network technique. Heliyon 9(4):e15455. https://doi.org/10.1016/j.heliyon.2023.e15455

Haberman D (2023) River of love in an age of pollution: the Yamuna river of northern India. Univ of California Press, California

Hafeez A, Shamair Z, Shezad N, Javed F, Fazal T, Rehman SU, Bazmi AA, Rehman F (2021) Solar powered decentralized water systems: a cleaner solution of the industrial wastewater treatment and clean drinking water supply challenges. J Clean Prod 289:125717. https://doi.org/10.1016/j.jclepro.2020.125717

Hassan MF, Sabri MA, Fazal H, Hafeez A, Shahzad N, Hussain M (2020) Recent trends in activated carbon fibers production from various precursors and applications—a comparative review. J Anal Appl Pyrolysis 145:104715. https://doi.org/10.1016/j.jaap.2019.104715

Hoseinian FS, Rezai B, Kowsari E, Safari M (2018) Kinetic study of Ni (II) removal using ion flotation: effect of chemical interactions. Miner Eng 119:212–221. https://doi.org/10.1016/j.mineng.2018.01.028

Hothi N, Hothi G (2022) Water crisis of Shimla town: past and present scenario. Mater Today: Proc 64:1250–1254. https://doi.org/10.1016/j.matpr.2022.03.715

Hou Q, Zhang Q, Huang G, Liu C, Zhang Y (2020) Elevated manganese concentrations in shallow groundwater of various aquifers in a rapidly urbanized delta, south China. Sci Total Environ 701:134777. https://doi.org/10.1016/j.scitotenv.2019.134777

Hussain SA, Sharma RK, Dasgupta N, Raha A (2011) Assessment of minimum water flow requirements of Chambal River in the context of Gharial (*Gavialis gangeticus*) and Gangetic dolphin (*Platanista gangetica*) conservation. Wildlife Institute of India

Isaac R, Siddiqui S (2022) Application of water quality index and multivariate statistical techniques for assessment of water quality around Yamuna River in Agra region, Uttar Pradesh, India. Water Sci Technol Water Supply 22(3):3399–3418. https://doi.org/10.2166/ws.2021.395

Jaiswal M, Gupta SK, Chabukdhara M, Nasr M, Nema AK, Hussain J, Malik T (2022) Heavy metal contamination in the complete stretch of Yamuna river: A fuzzy logic approach for comprehensive health risk assessment. PLoS ONE 17(8):0272562. https://doi.org/10.1371/journal.pone.0272562

Jaspal D, Malviya A (2020) Composites for wastewater purification: a review. Chemosphere 246:125788. https://doi.org/10.1016/j.chemosphere.2019.125788

Joshi SK, Swarnkar S, Shukla S, Kumar S, Jain S, Gautam S (2023) Snow/ice melt, precipitation, and groundwater contribute to the Sutlej river system. Water Air Soil Pollut 234(11):719. https://doi.org/10.1007/s11270-023-06744-4

Kashyap R, Verma KS, Bhardwaj SK, Sharma JK (2015) Hydrochemistry of dissolved metals in Yamuna River around industrial hub of Himachal Pradesh, India. Appl Biol Res 17(3):288–296. https://doi.org/10.5958/0974-4517.2015.00041.5

Kashyap R, Verma KS, Bhardwaj SK, Mahajan P, Sharma JK, Sharma R (2016) Water chemistry of Yamuna river along Ponta Sahib industrial hub of Himachal Pradesh, India. Res Environ Life Sci 9(3):277–281

Kaur S, Mehra P (2012) Assessment of heavy metals in summer & winter seasons in River Yamuna segment flowing through Delhi, India. J Environ Ecol. https://doi.org/10.5296/jee.v3i1.2675

Kaur L, Rishi MS, Sharma S, Sharma B, Lata R, Singh G (2019) Hydrogeochemical characterization of groundwater in alluvial plains of river Yamuna in northern India: an insight of controlling processes. J King Saud Univ Sci 31(4):1245–1253. https://doi.org/10.1016/j.jksus.2019.01.005

Kaur L, Rishi MS, Singh G, Nath Thakur S (2020) Groundwater potential assessment of an alluvial aquifer in Yamuna sub-basin (Panipat region) using remote sensing and GIS techniques in conjunction with analytical hierarchy process (AHP) and catastrophe theory (CT). Ecol Indic 110:105850. https://doi.org/10.1016/j.ecolind.2019.105850

Kaur HS, Warren A, Kamra K (2021) Spatial variation in ciliate communities with respect to water quality in the Delhi NCR stretch of River Yamuna, India. Eur J Protistol 79:125793. https://doi.org/10.1016/j.ejop.2021.125793

Kayastha AM (2024) Yamuna River's health: assessment for restoration. J Syst Sci Eng. 28(1):66

Khan A, Govil H, Taloor AK, Kumar G (2020) Identification of artificial groundwater recharge sites in parts of Yamuna River basin India based on remote sensing and geographical information system. Groundw Sustain Dev 11:100415. https://doi.org/10.1016/j.gsd.2020.100415

Khan AH, Aziz HA, Khan NA, Dhingra A, Ahmed S, Naushad M (2021) Effect of seasonal variation on the occurrences of high-risk pharmaceutical in drain-laden surface water: a risk analysis of Yamuna River. Sci Total Environ 794:148484. https://doi.org/10.1016/j.scitotenv.2021.148484

Khan MA, Khan N, Ahmad A, Kumar R, Singh A, Chaurasia D, Neogi S, Kumar V, Bhargava PC (2023) Potential health risk assessment, spatio-temporal hydrochemistry and groundwater quality of Yamuna river basin, Northern India. Chemosphere 311(1):136880. https://doi.org/10.1016/j.chemosphere.2022.136880

Khan MA (2022) Ravines and their afforestation along Chambal river in Madhya Pradesh

Khanna DR, Bhutiani R, Matta G, Singh V, Tyagi P, Tyagi B (2010) Water quality characteristics of river tons at district Dehradun, Uttarakhand (India). Environ Conserv J 11(1&2):119–123

Kluska M, Jabłońska J (2024) Pollution assessment and spatial distribution of heavy metals in surface waters and bottom sediments of the Krzna river (Poland). Water 16(7):1008. https://doi.org/10.3390/w16071008

Koop SHA, van Leeuwen CJ (2017) The challenges of water, waste and climate change in cities. Environ Dev Sustain 19:385–418. https://doi.org/10.1007/s10668-016-9760-4

Kr K, Sardar UR, Bhargavi E, Devi I, Bhunia B, Tiwari ON (2018) Advances in exopolysaccharides based bioremediation of heavy metals in soil and water: a critical review. Carbohydr Polym 199:353–365. https://doi.org/10.1016/j.carbpol.2018.07.037

Kumar A, Sharma MP, Yadav NS (2014) Assessment of water quality changes at two locations of Chambal River: M.P. J Mater Environ Sci 5:1781–1785

Kumar N, Hans S, Srivastava A (2021) Chromium bioremediation from river Yamuna using cyanobacteria at white and red illumination wavelengths. J Water Process Eng 44:102416. https://doi.org/10.1016/j.jwpe.2021.102416

Lamba M, Sreekrishnan TR, Ahammad SZ (2020) Sewage mediated transfer of antibiotic resistance to River Yamuna in Delhi. India J Environ Chem Eng 8(1):102088. https://doi.org/10.1016/j.jece.2017.12.041

Liu J, Mwamulima T, Wang Y, Fang Y, Song S, Peng C (2017) Removal of Pb (II) and Cr (VI) from aqueous solutions using the fly ash-based adsorbent material-supported zero-valent iron. J Mol Liq 243:205–211. https://doi.org/10.1016/j.molliq.2017.08.004

Madan R, Chaudhry S, Sharma M, Madan S (2018) Determination of water quality index of Indraprastha estate region and the vicinity area in Delhi, India. ESSENCE Int J Env Rehab Conserv 9(1):89–100. https://doi.org/10.31786/09756272.18.9.1.126

Malik D, Singh S, Thakur J, Singh RK, Kaur A, Nijhawan S (2014) Heavy metal pollution of the Yamuna River: an introspection. Int J Curr Microbiol App Sci 3(10):856–863

Mansor MI, Fatehah MO, Aziz HA, Wang LK (2024) Occurrence, behaviour and transport of heavy metals from industries in river catchments. Industrial waste engineering. Springer International Publishing, Cham, pp 205–277. https://doi.org/10.1007/978-3-031-46747-9_6

MapChart (2024) https://www.mapchart.net/india.html

Matta G, Bhutiani R, Singh V, Ishaq F (2012) Seasonal variation in physico-chemical characteristic status of River Yamuna in Doon Valley of Uttarakhand. Environ Conserv J 13(1–2):119–124. https://doi.org/10.36953/ECJ.2012.131222

Mayank P, Tyagi RK (2013) Studies on fish biodiversity and their conservation of the Yamuna river at Allahabad, Uttar Pradesh, India. J Kalash Sci 1:105–110

Mishra AK (2010) A river about to die: Yamuna. J Water Resour Prot 2:489. https://doi.org/10.4236/jwarp.2010.25056

Mishra S, Kumar A, Shukla P (2016) Study of water quality in Hindon River using pollution index and environmetrics, India. Desalin Water Treat 57:19121–19130. https://doi.org/10.1080/19443994.2015.1098570

Mishra S, Kumar P, Mehrotra I, Kumar M (2023) Prevalence of organic micropollutants in the Yamuna River, Delhi, India: seasonal variations and governing factors. Sci Total Environ 858(1):159684. https://doi.org/10.1016/j.scitotenv.2022.159684

Morali EK, Uzal N, Yetis U (2016) Ozonation pre and post-treatment of denim textile mill effluents: effects of cleaner production measures. J Clean Prod 137:1–9. https://doi.org/10.1016/j.jclepro.2016.07.059

Mrozik W, Rajaeifar MA, Heidrich O, Christensen P (2021) Environmental impacts, pollution sources and pathways of spent lithium-ion batteries. Energy Environ Sci 14:6099–6121. https://doi.org/10.1039/D1EE00691F

Muthaiyah NP (2020) Rejuvenating Yamuna River by wastewater treatment and management. Int J Energy Environ Sci. 5(1):14–29. https://doi.org/10.11648/J.IJEES.20200501.13

Naithani R, Pande IP (2015) Comparative analysis of the trends in river water quality parameters: a case study of the Yamuna River. Int J Sci 4(12):1212–1221

Naushad SS, Lall A, Charan AA (2014) Determination of heavy metals in water of Ganga and Yamuna river basin in Allahabad. Asian J Environ Sci 9(2):106–108. https://doi.org/10.15740/HAS/AJES/9.2/106-108

Nehra V, Singh SK (2020) Assessment of water quality of Najafgarh drain and its impact on River Yamuna. In: International conference of advance research & innovation

Nielsen GD, Søderberg U, Jørgensen PJ, Templeton DM, Rasmussen SN, Andersen KE, Grandjean P (1999) Adsorption and retention of nickel from drinking water in relation to food intake and nickel sensitivity. Toxicol Appl Pharmacol 154(1):67–75. https://doi.org/10.1006/taap.1998.8577

Nriagu J (2007) Zinc toxicity in humans. School of Public Health, University of Michigan. pp 1–7

Okpara EC, Fayemi OE, Wojuola OB, Onwudiwe DC, Ebenso EE (2022) Electrochemical detection of selected heavy metals in water: a case study of African experiences. RSC Adv 12:26319–26361. https://doi.org/10.1039/D2RA02733J

Omraei M, Esfandian H, Katal R, Ghorbani M (2011) Study of the removal of Zn(II) from aqueous solution using polypyrrole nanocomposite. Desalination 271:248–256. https://doi.org/10.1016/j.desal.2010.12.038

Pal R, Dubey RK, Dubey SK, Singh AK (2017) Assessment of heavy metal pollution through index analysis for Yamuna water in Agra region, India. Int J Curr Microbiol App Sci 6(12):1491–1498. https://doi.org/10.20546/ijcmas.2017.612.166

Pandey N, Kumar P, Ali S, Vishwakarma BK, Kumar S (2018) Role of small tributaries in ichthyofaunal diversity of rivers in Uttarakhand. J Coldwater Fish 1(1):89–96

Parihar K, Sankhla MS, Kumar R (2019) Water quality status of Yamuna River and its toxic effects on humans. Environ Anal Eco Stud 6(1):000628. https://doi.org/10.31031/EAES.2019.06.000628

Parween M, Al R, Raju NJ (2017) Waste water management and water quality of river Yamuna in the megacity of Delhi. Int J Environ Sci Technol 14:2109–2124. https://doi.org/10.1007/s13762-017-1280-8

Parween M, Ramanathan AL, Raju NJ (2021) Assessment of toxicity and potential health risk from persistent pesticides and heavy metals along the Delhi stretch of river Yamuna. Environ Res 202:111780. https://doi.org/10.1016/j.envres.2021.111780

Patel PP, Mondal S, Ghosh KG (2020) Some respite for India's dirtiest river? Examining the Yamuna's water quality at Delhi during the COVID-19 lockdown period. Sci Total Environ 744:140851. https://doi.org/10.1016/j.scitotenv.2020.140851

Pattnayak KC, Awasthi A, Sharma K, Pattnayak BB (2023) Fate of rainfall over the North Indian states in the 15 and 2 °C warming scenarios. Earth Space Sci. 10(2):002671. https://doi.org/10.1029/2022ea002671

QGIS. https://www.qgis.org/en/site/

Rai RK, Upadhyay A, Ojha CSP, Singh VP (2012) Salient features of the Yamuna River basin. The Yamuna River basin: water resources and environment. Springer, Dordrecht, pp 13–25. https://doi.org/10.1007/978-94-007-2001-5_2

Rajan S, Janardhana Raju N (2023) Heavy metal contamination in surface and groundwater and its human health risk assessment in the upper Yamuna River basin, India. In: EGU general assembly conference abstracts. pp. EGU-704. https://ui.adsabs.harvard.edu/link_gateway/2023EGUGA..25..704R/doi:10.5194/egusphere-egu23-704

Rajan S, Nandimandalam JR (2024) Environmental health risk assessment and source apportion of heavy metals using chemometrics and pollution indices in the upper Yamuna River basin, India. Chemosphere 346:140570. https://doi.org/10.1016/j.chemosphere.2023.140570

Ravindra K, Kaushik A (2003) Seasonal variations in physico-chemical characteristics of River Yamuna in Haryana and its ecological best-designated use. J Environ Monit 5:419–426. https://doi.org/10.1039/B301723K

Reddy IS, David AA, Srinidhi P (2021) Anatomization of irrigation water quality parameters of Chaka block, Yamuna river bank, Prayagraj, Uttar Pradesh India. Int J Chem Stud 9(2):36–40. https://doi.org/10.22271/chemi.2021.v9.i2a.11903

Saini G, Kumar A, Saini RP (2020) Assessment of hydrokinetic energy - a case study of eastern Yamuna canal. Mater Today: Proc 46(2):5223–5227. https://doi.org/10.1016/j.matpr.2020.08.595

Saksena DN, Garg RK, Rao RJ (2008) Water quality and pollution status of Chambal river in National Chambal sanctuary, Madhya Pradesh. J Environ Biol 29(5):701–710

Samson R, Shah M, Yadav R, Sarode P, Rajput V, Dastager SG, Dharne MS, Khairnar K (2019) Metagenomic insights to understand transient influence of Yamuna River on taxonomic

and functional aspects of bacterial and archaeal communities of River Ganges. Sci Total Environ 674:288–299. https://doi.org/10.1016/j.scitotenv.2019.04.166

Sankhla MS, Kumari M, Nandan M, Kumar R, Agrawal P (2016) Heavy metals contamination in water and their hazardous effect on human health-a review. Int J Curr Microbiol App Sci 5(10):759–766. https://doi.org/10.20546/ijcmas.2016.510.082

Sankhla MS, Kumar R, Prasad L (2022) Impact of variation in climatic changes in concentration of lead & nickel in Yamuna River water, Delhi, India. Mater Today: Proc 69(4):1540–1547. https://doi.org/10.1016/j.matpr.2022.05.242

Sarkar A, Shekhar S (2018) Iron contamination in the waters of upper Yamuna basin. Groundw Sustain Dev 7:421–429. https://doi.org/10.1016/j.gsd.2017.12.011

Sarkar S, Banerjee A, Halder U, Biswas R, Bandopadhyay R (2017) Degradation of synthetic azo dyes of textile industry: a sustainable approach using microbial enzymes. Water Conserv Sci Eng 2:121–131. https://doi.org/10.1007/s41101-017-0031-5

Sarkar AM, Rahman AKML, Samad A, Bhowmick AC, Islam JB (2019) Surface and ground water pollution in Bangladesh: a review. Asian Rev Environ Earth Sci 6(1):47–69. https://doi.org/10.20448/journal.506.2019.61.47.69

Sarker B, Keya KN, Mahir FI, Nahiun KM, Shahida S, Khan RA (2021) Surface and ground water pollution: causes and effects of urbanization and industrialization in South Asia. Sci Rev 7(3):32–41. https://doi.org/10.32861/sr.73.32.41

Sehgal M, Garg A, Suresh R, Dagar P (2012) Heavy metal contamination in the Delhi segment of Yamuna basin. Environ Monit Assess 184(2):1181–1196. https://doi.org/10.1007/s10661-011-2031-9

Selvaraj S, Krishnaswamy S, Devashya V, Sethuraman S, Krishnan UM (2011) Investigations on membrane perturbation by Chrysin and its copper complex using self-assembled lipid bilayers. Langmuir 27(21):13374–13382. https://doi.org/10.1021/la2029356

Semwal N, Akolkar P (2006) Water quality assessment of sacred Himalayan rivers of Uttaranchal. Curr Sci 91(4):486–496

Sharma K (2015) Pollution study of River Yamuna: the Delhi story. Int J Sci Res 6(10):1718–1722

Sharma S, Gupta AS (2022) Impact of biological parameters on water quality in Himalayan and upper segments of River Yamuna. Eco Environ Conserv 28:284–287. https://doi.org/10.53550/EEC.2022.v28i06s.0048

Sharma P, Meher PK, Kumar A, Gautam YP, Mishra KP (2014) Changes in water quality index of Ganges river at different locations in Allahabad. Sustain Water Qual Ecol 3–4:67–76. https://doi.org/10.1016/j.swaqe.2014.10.002

Sharma M, Kansal A, Jain S, Sharma P (2015) Application of multivariate statistical techniques in determining the spatial temporal water quality variation of Ganga and Yamuna Rivers present in Uttarakhand state, India. Water Qual Expo Health 7(4):567–581. https://doi.org/10.1007/s12403-015-0173-7

Sharma R, Kumar R, Satapathy SC, Ansari NA, Singh KK, Mahapatra RP, Agarwal AK, Le HV, Pham BT (2020a) Analysis of water pollution using different physicochemical parameters: a study of Yamuna River. Front Environ Sci 8:581591. https://doi.org/10.3389/fenvs.2020.581591

Sharma VK, Jinadatha C, Lichtfouse E (2020b) Environmental chemistry is most relevant to study coronavirus pandemics. Environ Chem Lett 18:993–996. https://doi.org/10.1007/s10311-020-01017-6

Sharma R, Kumar A, Singh N, Sharma K (2021) Impact of seasonal variation on water quality of Hindon River: physicochemical and biological analysis. SN Appl Sci 3:28. https://doi.org/10.1007/s42452-020-03986-3

Sharma VR, Bisht K, Sanu SK (2022) Assessment of physicochemical characteristics of water quality along Balua Ghat, Yamuna River in Prayagraj Metropolitan City, India. Indian J Geogr 19:45–54

Sharma M, Chaudhry S (2015) Impact of industrial pollution on Yamuna River: a review. https://doi.org/10.13140/RG.2.1.3632.8401

Sharma D, Kansal A (2011) Current condition of the Yamuna River: an overview of flow, pollution load, and human use. Yamuna River: A confluence of waters, a crisis of need. p 17

Shehzad N, Zafar M, Ashfaq M, Razzaq A, Akhter P, Ahmad N, Hafeez A, Azam K, Hussain M, Kim WY (2020) Development of AgFeO2/rGO/TiO2 ternary composite photocatalysts for enhanced photocatalytic dye decolorization. Crystals 10(10):923. https://doi.org/10.3390/cryst10100923

Singh G, Patel N, Jindal T, Srivastava P, Bhowmik A (2020) Assessment of spatial and temporal variations in water quality by the application of multivariate statistical methods in the Kali River, Uttar Pradesh. India Environ Monit Assess 192:394. https://doi.org/10.1007/s10661-020-08307-0

Sinha S (2023) Correlative assessment of water quality and qualitative and quantitative fish production from River Yamuna, In the state NCT Delhi. Bullet Pure Appl Sci Zoology. https://doi.org/10.48165/bpas.2023.42A.1.10

Sisodiya S, Mathur AK (2021) An assessment of water quality indices of Chambal river in Kota city for drinking and irrigation purposes. WEENTECH Proc Energy. https://doi.org/10.32438/wpe.122021

Srivastava NK, Majumder CB (2008) Novel biofiltration methods for the treatment of heavy metals from industrial wastewater. J Hazard Mater 151(1):1–8. https://doi.org/10.1016/j.jhazmat.2007.09.101

Syeed MMM, Hossain MS, Karim MR, Uddin MF, Hasan M, Khan RH (2023) Surface water quality profiling using the water quality index, pollution index and statistical methods: a critical review. Environ Sustain Indic 18:100247. https://doi.org/10.1016/j.indic.2023.100247

Thakur N, Rishi M, Sharma DA, Keesari T (2018) Quality of water resources in Kullu Valley in Himachal Himalayas, India: perspective and prognosis. Appl Water Sci 8:20. https://doi.org/10.1007/s13201-018-0668-z

Thakur D, Sharma A, Goel P, Thakur A, Raturi M (2023) Groundwater quality assessment in the alluvial region of upper Yamuna basin. India Groundw Sustain Dev 22:100969. https://doi.org/10.1016/j.gsd.2023.100969

Tripathi A, Rajwar GS, Sharma RC (2008) Impacts of industrial effluents on Asan river, Doon. Valley Int J Chem Sci 6(4):2155–2171

Upadhyay A, Rai RK, Upadhyay A, Rai RK (2013) Brief overview of the Yamuna river basin and issues. Water management and public participation: case studies from the Yamuna River basin, India. Springer, Dordrecht, pp 13–24

Usha K, Singh B (2024) Reifying "Yamuna": unpacking the pluriversal possibilities for rejuvenation of the river at Poiaghat, Swarg Dhaam. Agra. J Syst Sci Eng 28(1):40

Vasistha P, Ganguly R (2020) Water quality assessment of natural lakes and its importance: an overview. Mater Today: Proc 32(4):544–552. https://doi.org/10.1016/j.matpr.2020.02.092

Verma N, Singh G, Ahsan N (2022) Development of water quality management strategies for an urban river reach: a case study of the river Yamuna, Delhi. India Arab J Geosci 15:1767. https://doi.org/10.1007/s12517-022-11030-4

Vishwakarma S, Varma A, Saxena G (2013) Assessment of water quality of Betwa River, Madhya Pradesh, India. Int J Water Resour 5(4):217–222

Vrat P (2024) Need for an Integrated Systems Approach in Rejuvenating the River Yamuna. J Syst Sci Eng 28(1): 5–7

Water Quality Data: Uttarakhand Pollution Control Board, Government Of Uttarakhand, India [WWW Document] (2022). URL https://ueppcb.uk.gov.in/pages/display/96-water-quality-data (Accessed 4 May 2023)

WHO (2011) Guidelines for drinking water quality, 4th edn. International Standard for Drinking Water Guidelines for Water Quality, Geneva, Switzerland

Wu G, Hong J, Li D, Wu Z (2019) Efficiency assessment of pollutants discharged in urban wastewater treatment: evidence from 68 key cities in China. J Clean Prod 233:1437–1450. https://doi.org/10.1016/j.jclepro.2019.06.012

Yadav A, Khandegar V (2019) Dataset on assessment of River Yamuna, Delhi, India using indexing approach. Data Brief 22:1–10. https://doi.org/10.1016/j.dib.2018.11.130

Yadav M, Yadav E (2024) The pollution status of some North Indian rivers: a review. J Exp Zool India. https://doi.org/10.51470/jez.2024.27.1.1

Yadav NS, Kumar A, Pani S, Sharma MP (2014) Water quality assessment of Chambal River in National Chambal Sanctuary of Madhya Pradesh. In: Nikil J (ed) Ecological sustainability: concept, principle, evidences and innovations. Excellent Publishing House, New Delhi, pp 44–52

Yadav N, Rajendra K, Awasthi A, Singh C (2023) Systematic exploration of heat wave impact on mortality and urban heat island: A review from 2000 to 2022. Urban Clim 51:101622. https://doi.org/10.1016/j.uclim.2023.101622

Zehra R, Singh SP, Verma J, Kulshreshtha A (2023) Spatio-temporal investigation of physico-chemical water quality parameters based on comparative assessment of QUAL 2Kw and WASP model for the upper reaches of Yamuna River stretching from Paonta Sahib, Sirmaur district to Cullackpur, North Delhi districts of North India. Environ Monit Assess 195(4):480. https://doi.org/10.1007/s10661-023-11072-5

Zhang X, Wang X (2015) Adsorption and desorption of Nickel(II) ions from aqueous solution by a lignocellulose/montmorillonite nanocomposite. PLoS ONE 10:0117077. https://doi.org/10.1371/journal.pone.0117077

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Authors and Affiliations

Madhuben Sharma[1] · Sameeksha Rawat[1] · Dheeraj Kumar[1] · Amit Awasthi[2] · Abhijit Sarkar[3] · Atul Sidola[4] · Tanupriya Choudhury[5,6] · Ketan Kotecha[6,7]

✉ Amit Awasthi
awasthitiet@gmail.com

✉ Tanupriya Choudhury
tanupriya@ddn.upes.ac.in;tanupriya.choudhury@sitpune.edu.in;
tanupriya1986@gmail.com

Madhuben Sharma
mbsharma@ddn.upes.ac.in

Sameeksha Rawat
sameeksha712rawat@gmail.com

Dheeraj Kumar
kdhiraj865@gmail.com

Abhijit Sarkar
abhijitbhu@gmail.com

Atul Sidola
asidola@gmail.com

Ketan Kotecha
director@sitpune.edu.in

1 Sustainability Cluster, University of Petroleum and Energy Studies, Dehradun, Uttarakhand, India 248007

2 Department of Physics, School of Engineering, University of Petroleum and Energy Studies, Dehradun, Uttarakhand, India 248007

3 Laboratory of Applied Stress Biology, Department of Botany, University of Gour Banga, Malda, West Bengal, India

4 Uurja Foundation, Dehradun, Uttarakhand, India

5 Present Address: School of Computer Sciences, University of Petroleum and Energy Studies (UPES), Bidholi Energy Acres Campus, Dehradun, Uttarakhand 248007, India

6 Present Address: Symbiosis Centre for Applied Artificial Intelligence, Symbiosis Institute of Technology, Symbiosis International (Deemed University), Pune, Maharashtra 411045, India

7 Peoples Friendship University of Russia named after Patrice Lumumba (RUDN University), 6 Miklukho-Maklaya Str., Moscow 117198, Russian Federation